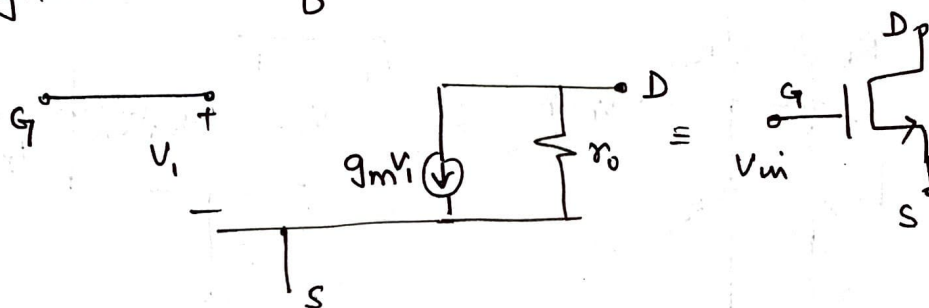


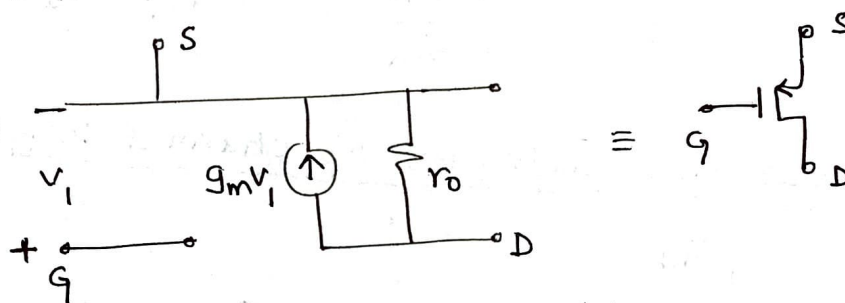
(1)

Common Source stage:-

Small signal model of n-MOS :-

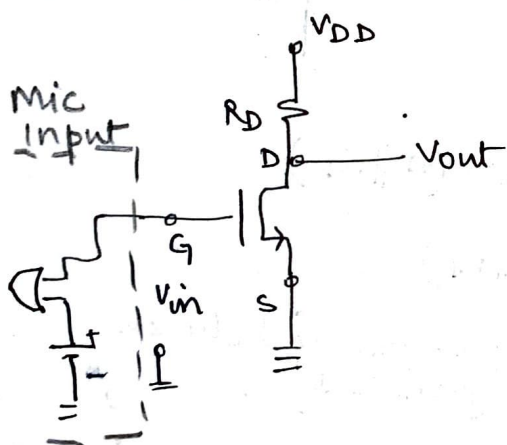


for p-MOS Devices :-

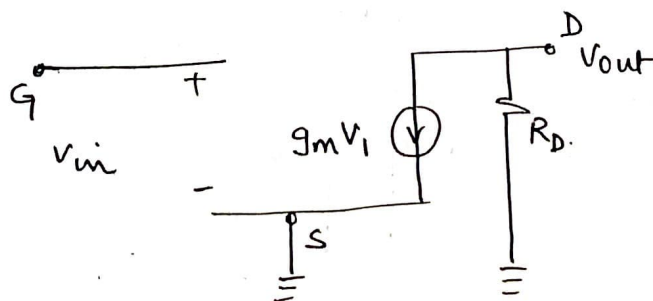


Voltage gain calculations in MOS (small-signal):-

find out $\frac{V_{out}}{V_{in}}$ (ac gain).



Small signal reduction:-



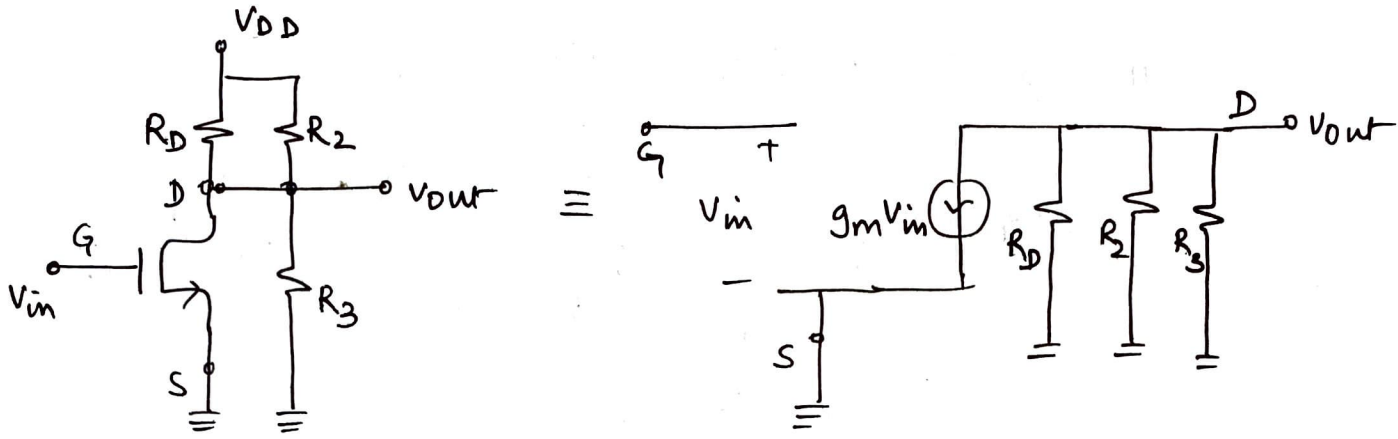
$$V_{out} = -R_D \cdot g_m V_i$$

$$\text{But } V_i = V_{in}$$

$$\therefore \frac{V_{out}}{V_{in}} = -g_m R_D$$

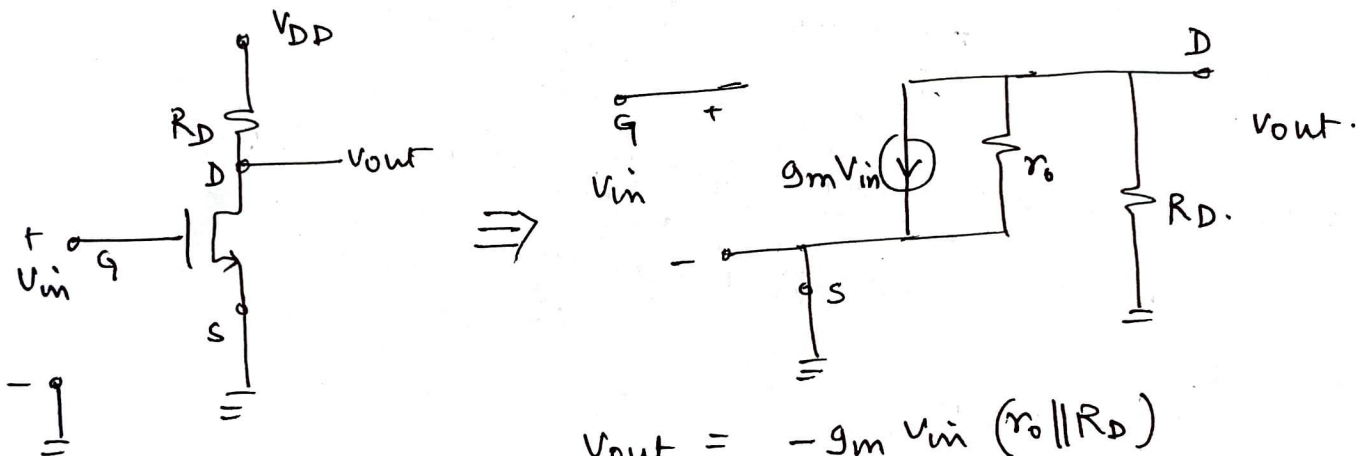
Note gain = $-g_m \times [\text{total resistance tied from drain to ac ground}]$

Say, add some more resistance :-



$$\therefore \frac{V_{out}}{V_{in}} = -g_m (R_D \parallel R_2 \parallel R_3)$$

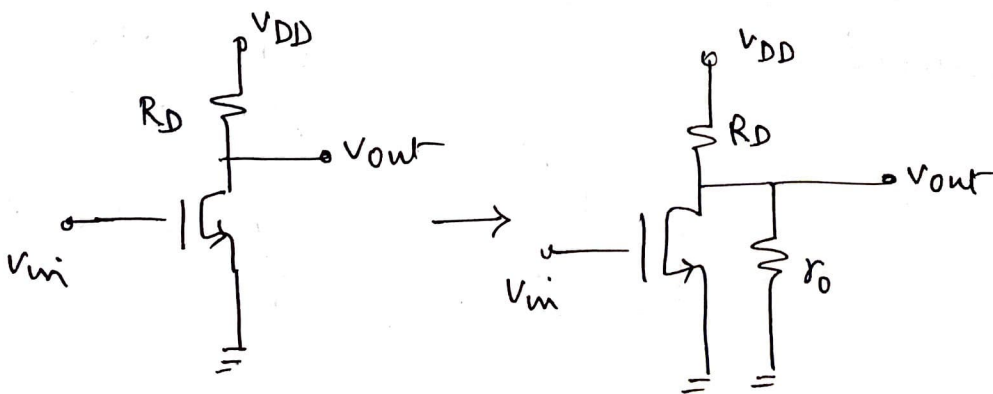
Gain under the inclusion of channel length modulation :-



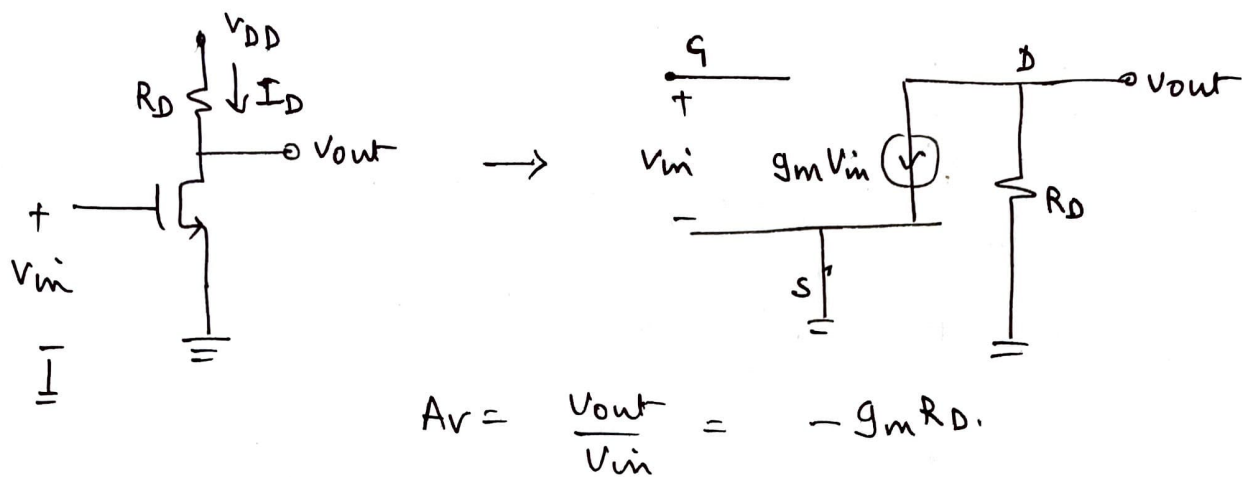
$$V_{out} = -g_m V_{in} (r_o \parallel R_D)$$

$$\therefore \frac{V_{out}}{V_{in}} = -g_m (r_o \parallel R_D)$$

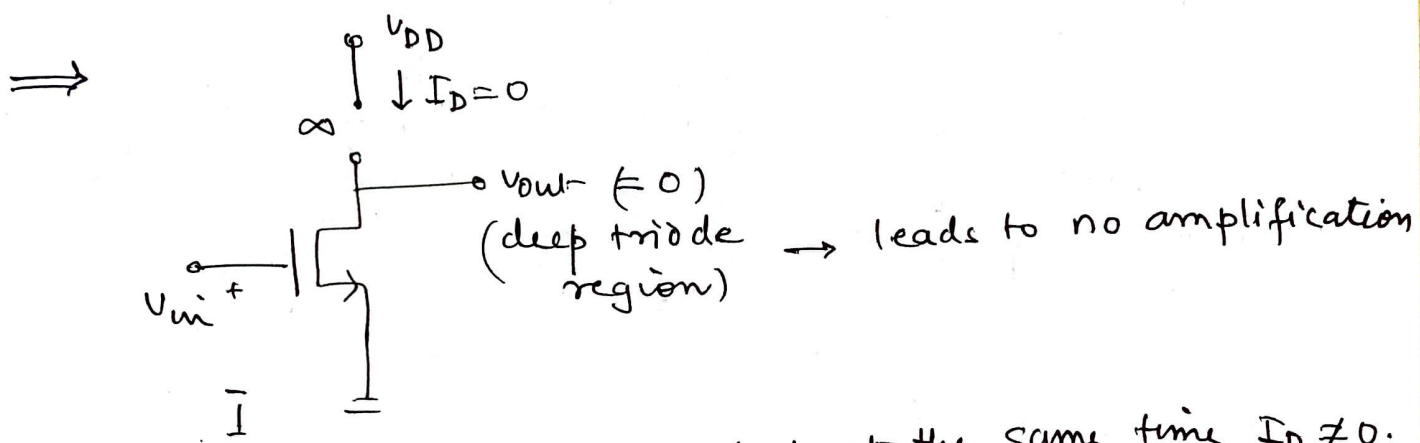
$$= -g_m [\text{total resistance between ac ground \& D}]$$



How to maximize the gain?

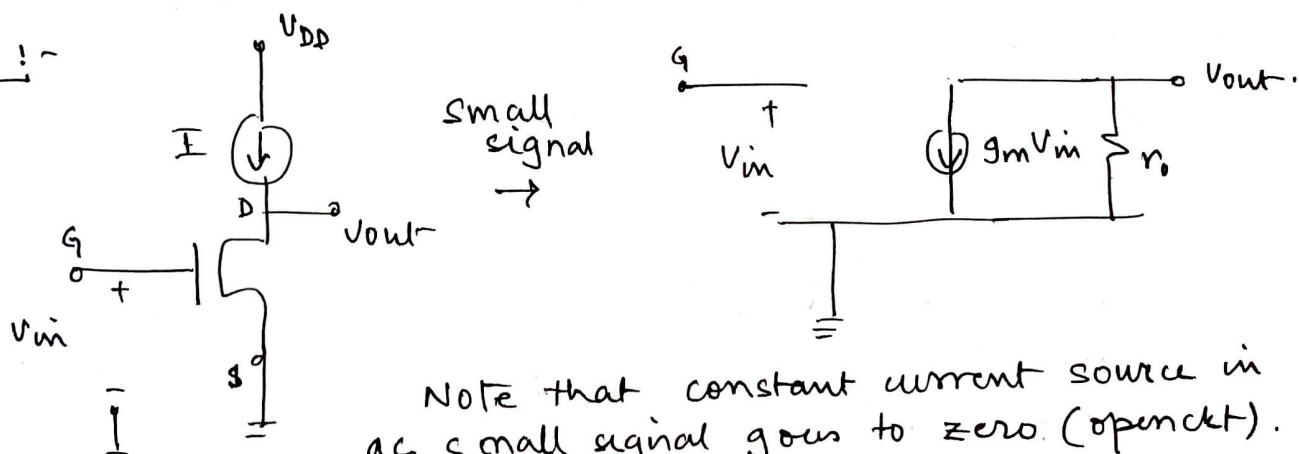


To increase A_v one can think $R_D \rightarrow \infty$, but for $R_D \rightarrow \infty$, $I_D \rightarrow 0$.



$\therefore$ One needs to have an $R_D \rightarrow \infty$ but at the same time $I_D \neq 0$.
what to do?

Do this!:-



Note that constant current source in ac small signal goes to zero (openckt).
This resemblance $R_D \rightarrow \infty$, but technically there is a current $I_D \neq 0$.

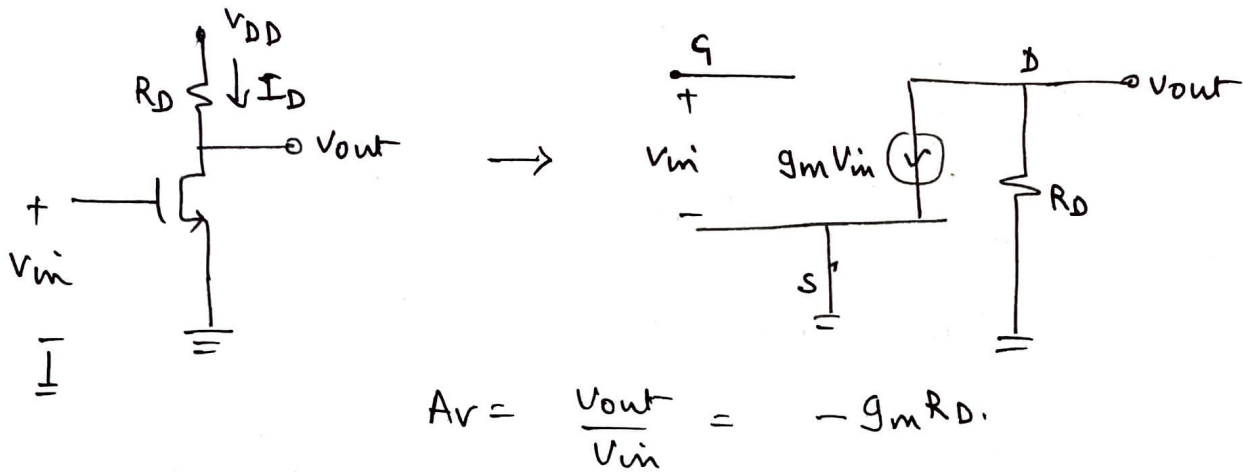
So maximum gain :-

$$A_v = -g_m r_o.$$

$\rightarrow$ intrinsic gain of MOS device.

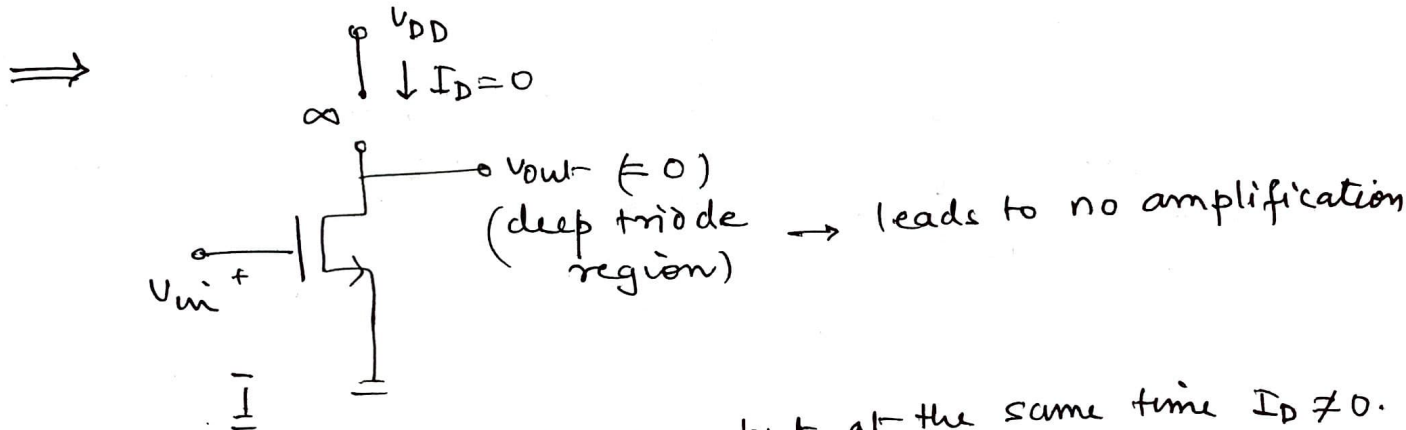
How to maximize the gain?

(3)



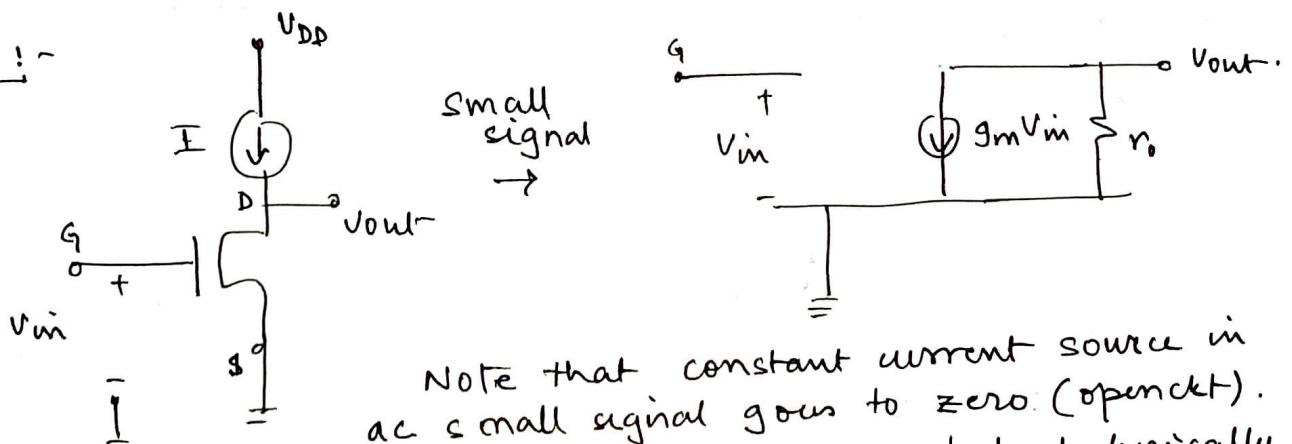
$$A_v = \frac{V_{out}}{V_{in}} = -g_m R_D.$$

To increase A_v one can think $R_D \rightarrow \infty$, but for $R_D \rightarrow \infty$, $I_D \rightarrow 0$.



$\therefore$ One needs to have an $R_D \rightarrow \infty$ but at the same time $I_D \neq 0$.
What to do?

Do this! -



Note that constant current source in ac small signal goes to zero (openckt).
This resembles $R_D \rightarrow \infty$, but technically there is a current $I_D \neq 0$.

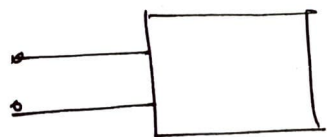
So maximum gain :-

$$A_v = -g_m r_o.$$

$\rightarrow$ intrinsic gain of MOS device.

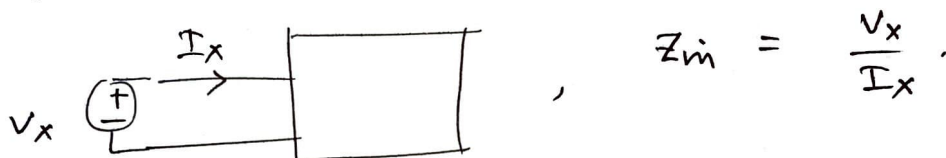
Concept of Impedance :-

(4)

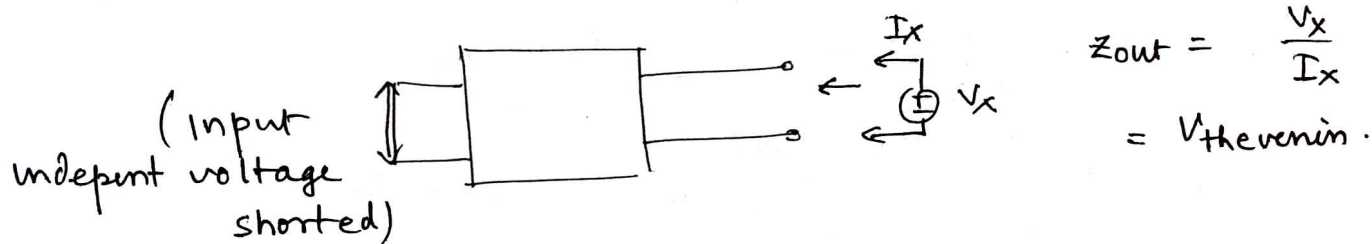


① Put all independent sources to zero. (voltage: s.c. current: o'ckt).

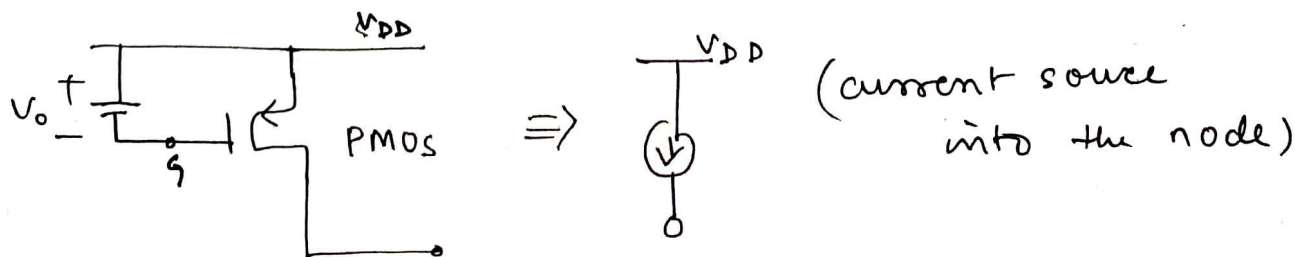
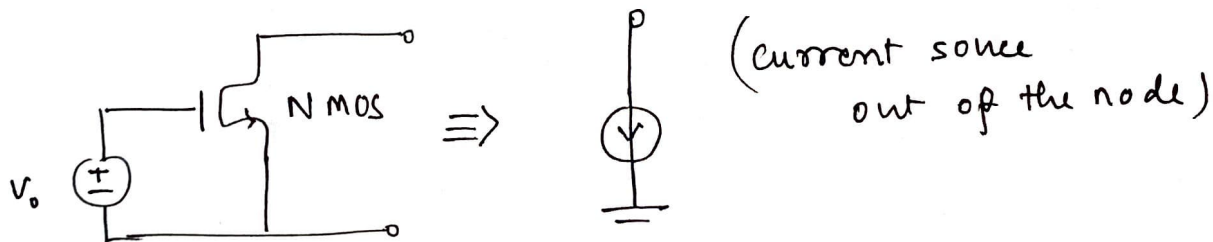
② Apply a small signal voltage V_x and measure I_x .



Output Impedance :-

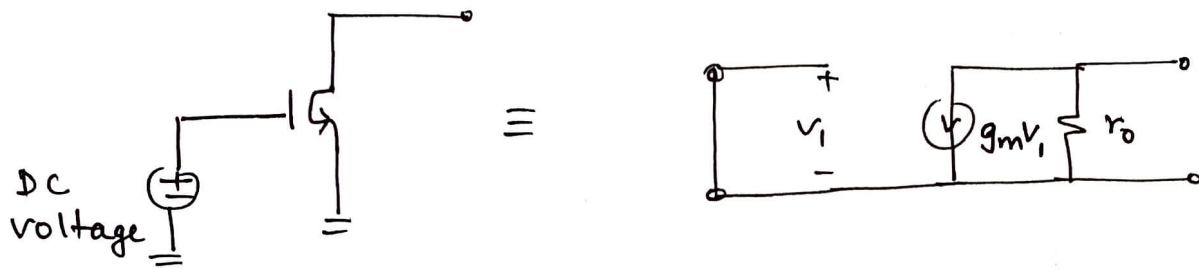


Current sources :-

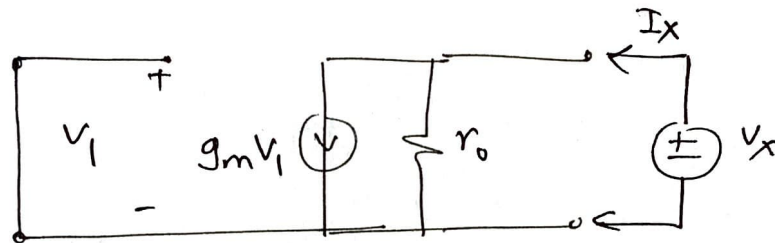


Small signal model of n-MOS current source:

5

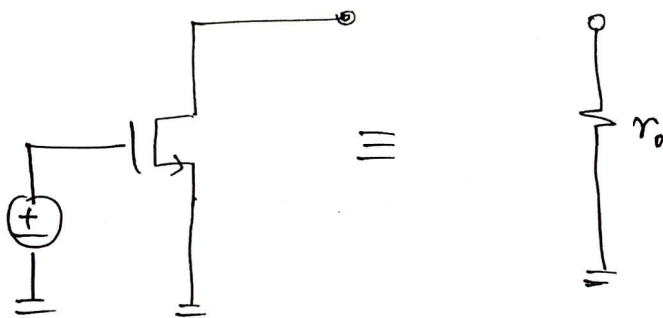


Impedance from outside :



Here $v_i = 0$ (short ckted). $\therefore g_m v_i = 0$.

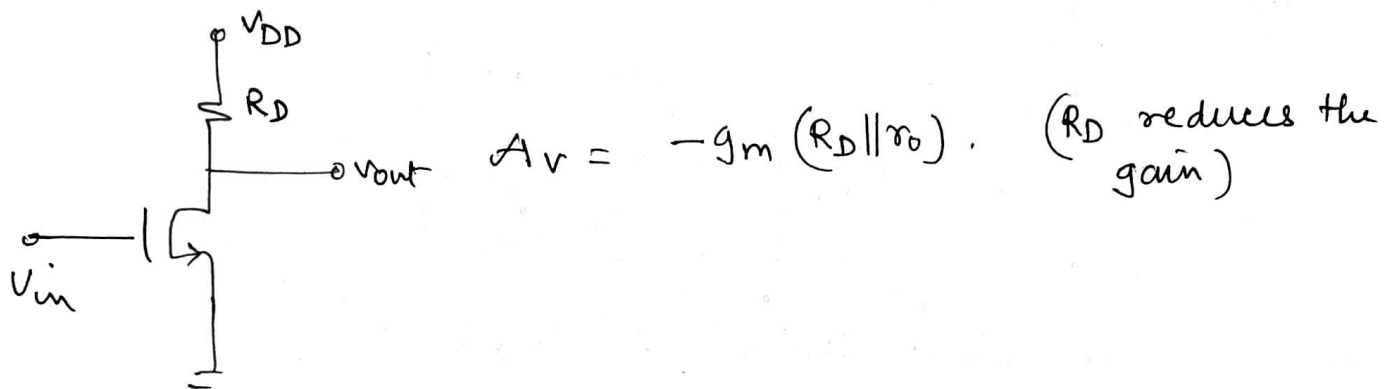
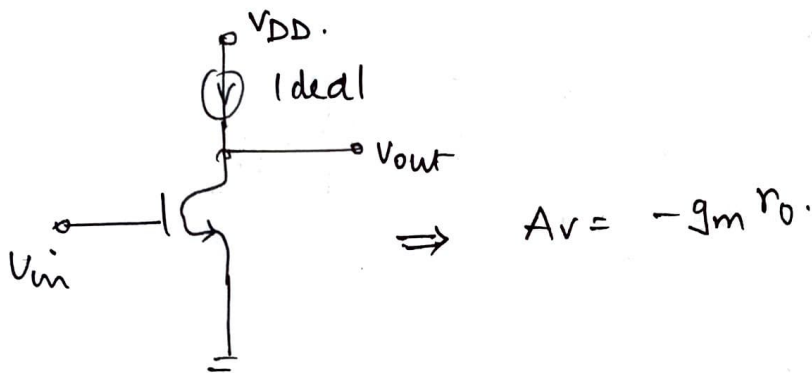
$$\therefore \frac{v_x}{I_x} = r_o$$



$\therefore$ Replace a MOS device acting as a current source by r_o .

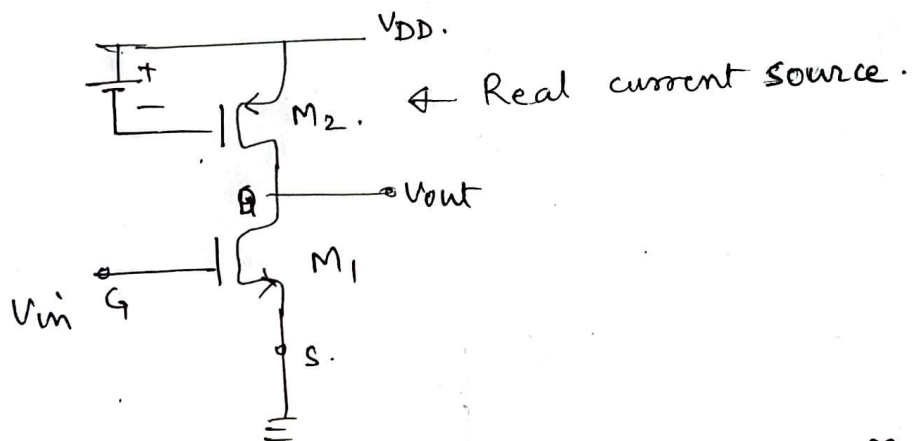
Common source stage with current source load:-

(6)

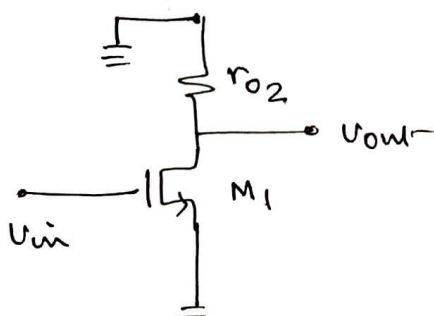


Now, note that the current source is pouring current into the node (device).

$\therefore$ we take the p-MOS as current source:



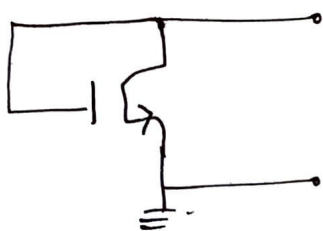
$\therefore$ Small signal analysis:- Replace c.s with r_o to ac ground)



$$\begin{aligned}
 A_v &= -g_{m1} \times (r_{o1} || r_{o2}) \\
 &= g_m \text{ of amplifying device} \\
 &\times \text{sum of resistance from drain to ac. ground.}
 \end{aligned}$$

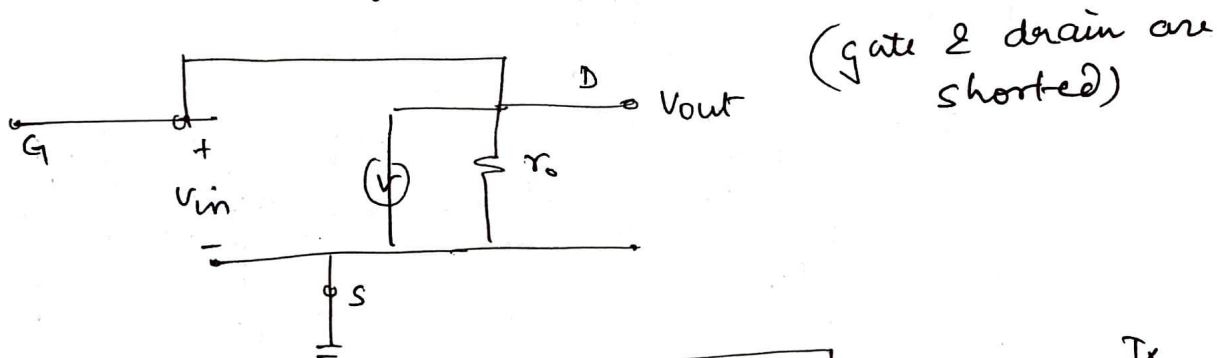
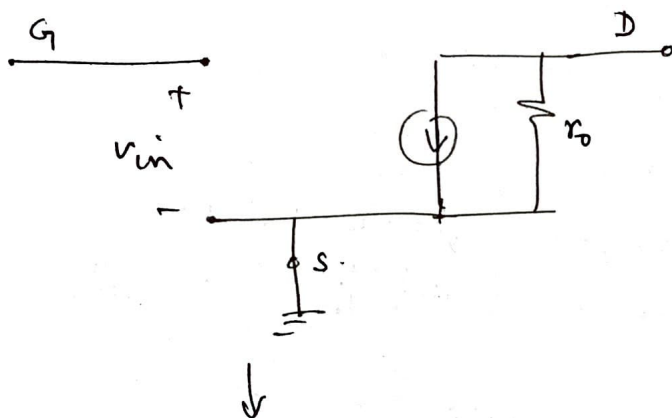
Diode connected MOS device.

(7)



Two terminal device.

Small signal model :-



Output impedance :-

$$R_{out} = \frac{V_x}{I_x}$$

Here $V_1 = V_x = V_{in}$

$$I_x = \frac{V_x}{r_o} + g_m V_1 = \frac{V_x}{r_o} + g_m V_x$$

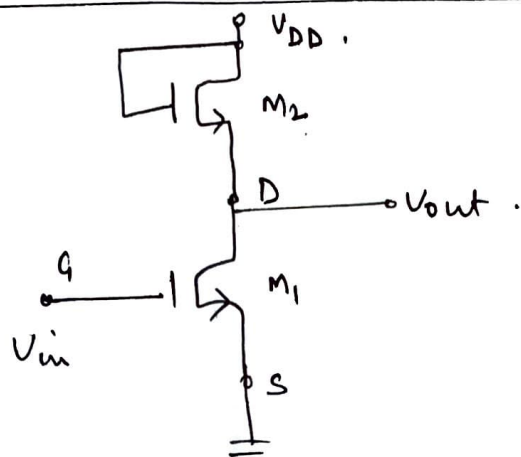
$$\therefore I_x = V_x \left(g_m + \frac{1}{r_o} \right)$$

$$\therefore \frac{V_x}{I_x} = \frac{1}{g_m + \frac{1}{r_o}} = \frac{r_o}{1 + r_o g_m} = r_o \parallel \frac{1}{g_m}$$

for diode connected device, $R_{out} = \frac{1}{g_m}$ for $r_o \rightarrow \infty$.
for nMOS current source, $R_{out} = r_o$.

CS stage with diode connected load:

(8)



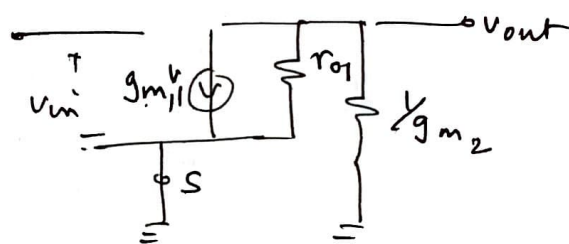
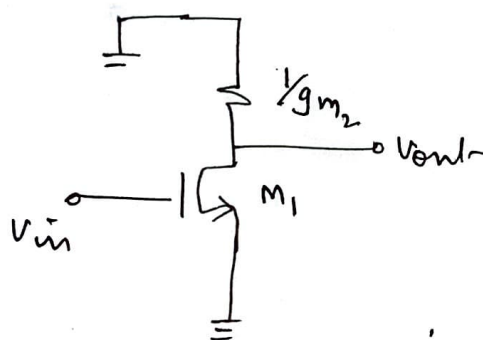
What is the voltage gain!?

Let say $\lambda = 0$ first,

then diode connected source load :-

$$R_{out} = r_o \parallel \frac{1}{g_m} = \frac{1}{g_m} \quad (\because r_o \rightarrow \infty)$$

$\therefore$ small signal analysis:



$$\therefore V_{out} = -g_m V_i (r_{o1} \parallel \frac{1}{g_{m2}})$$

$$\therefore A_v = -g_{m1} (r_{o1} \parallel \frac{1}{g_{m2}})$$

$$= -\frac{g_{m1}}{g_{m2}} \quad (\text{for } r_{o1} \rightarrow \infty)$$

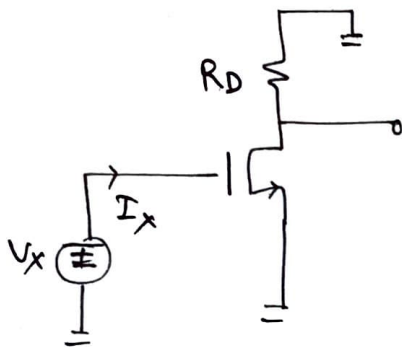
$$A_v = -\frac{g_{m1}}{g_{m2}} = -\frac{\sqrt{\mu_n C_{ox} (W/L)_1 I_{D1}}}{\sqrt{\mu_n C_{ox} (W/L)_2 I_{D2}}} = -\frac{\sqrt{(W/L)_1}}{\sqrt{(W/L)_2}}$$

only depends of geometry and temperature independent

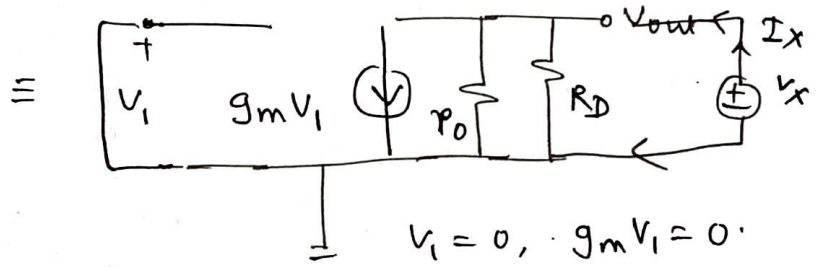
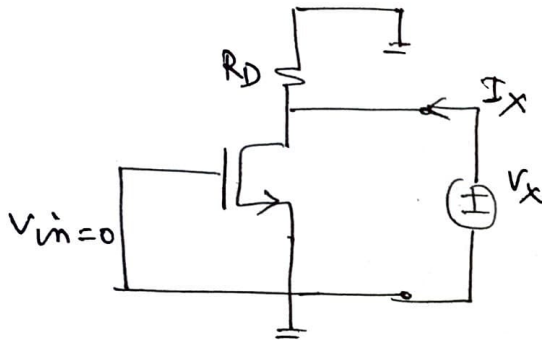
$$\lambda \neq 0, \quad A_v = -g_{m1} \cdot r_{o1} \parallel \left(r_{o2} \parallel \frac{1}{g_{m2}} \right)$$

Input and output Impedances:-

(9)

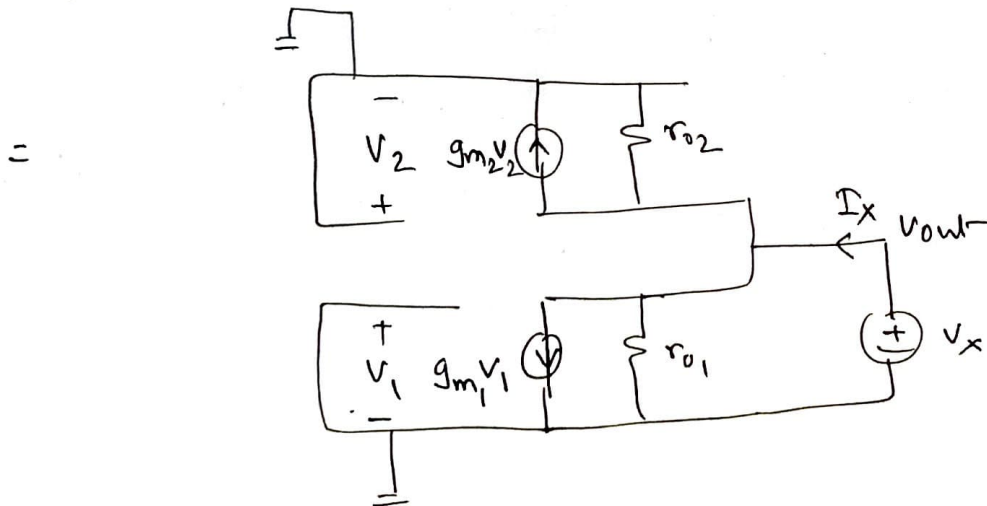
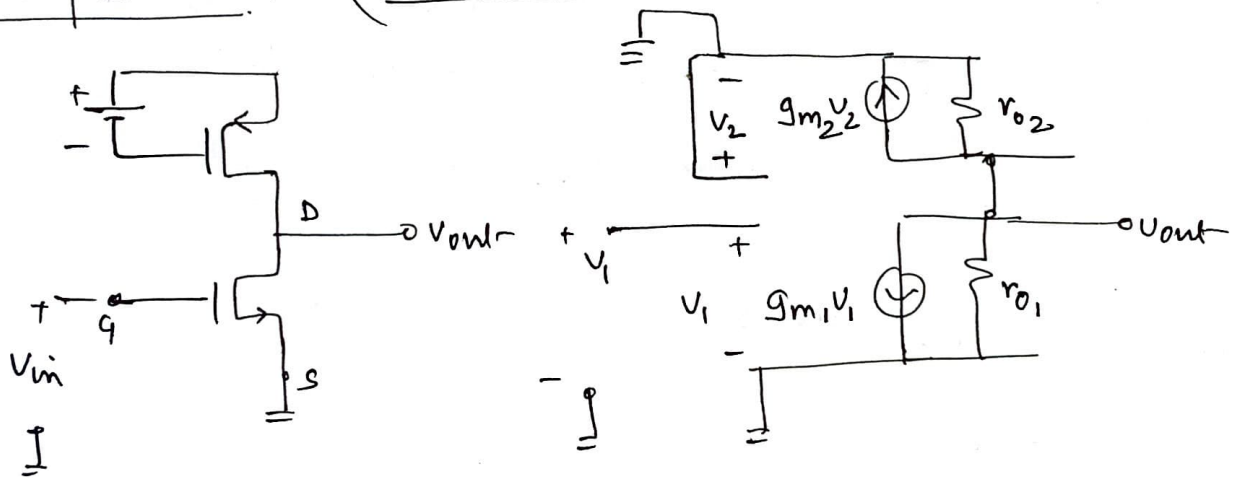


Input impedance:- $\frac{V_x}{I_x} = \infty$.



$\therefore \frac{V_x}{I_x} = R_D \parallel r_o \therefore R_{out} = r_o \parallel R_D$

I/O Impedance:- (Current source load):



$V_2 = 0, V_1 = 0$

$g_{m1}V_1 = g_{m2}V_2 = 0$

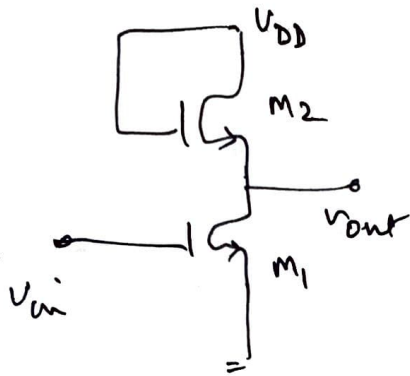
$\therefore \frac{V_x}{I_x} = (r_{o1} \parallel r_{o2})$

$\therefore R_{out} = (r_{o1} \parallel r_{o2})$

$R_{in} = \infty$

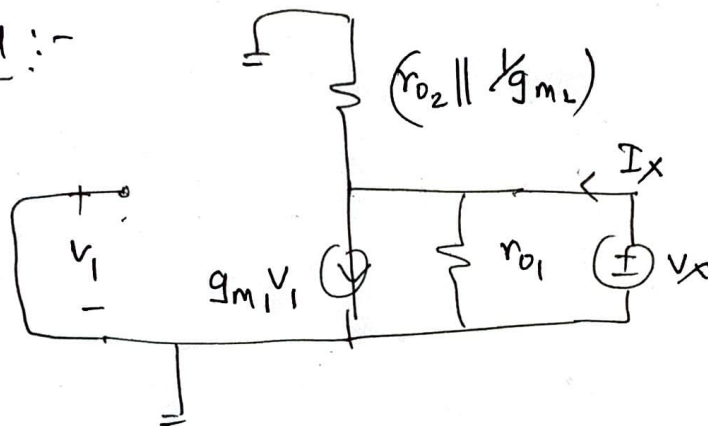
CS stage diode connected source

(10)



Small signal :-

$R_{out}?$

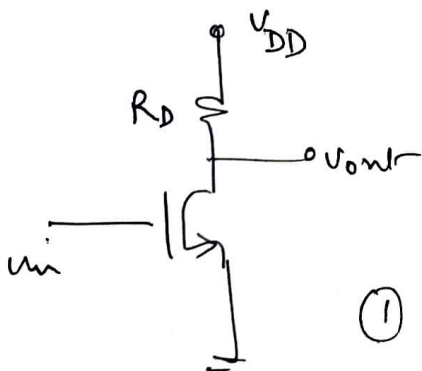


Here $v_1 = 0$. $\therefore g_{m1} v_1 = 0$ $\therefore \frac{v_x}{I_x} = r_{o1} \parallel (r_{o2} \parallel 1/g_{m2})$

Note that $1/g_{m2}$ is smaller than r_{o1} & r_{o2} .

$\therefore R_{out} \sim 1/g_{m2}$

factors affecting gain :-



$$A_v = -g_m R_D$$

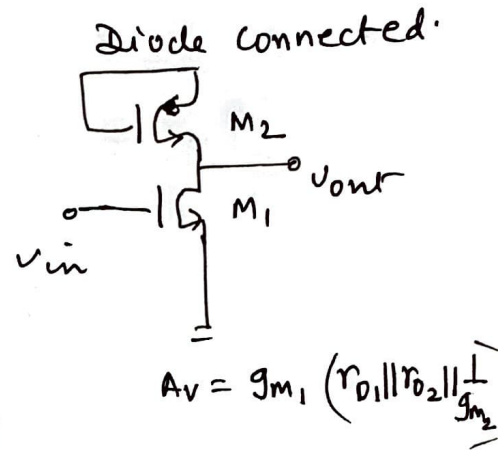
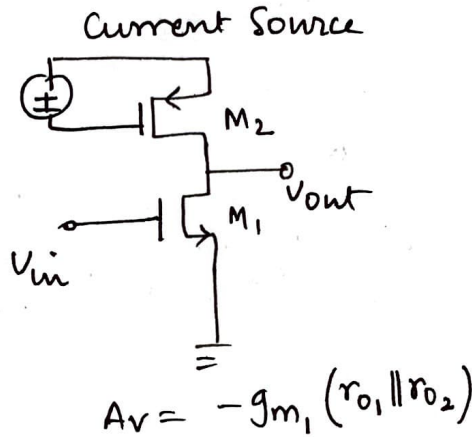
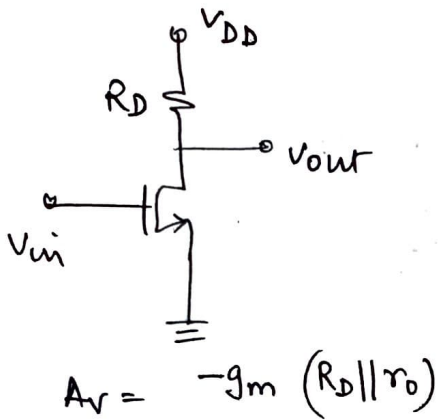
$$g_m = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right) I_D}$$

- ① Prone to temperature. $0^\circ\text{C} \rightarrow 80^\circ\text{C}$.
- ② Supply voltage. (I_D depends on that)
- ③ Process variations.
- ④ Signal level.

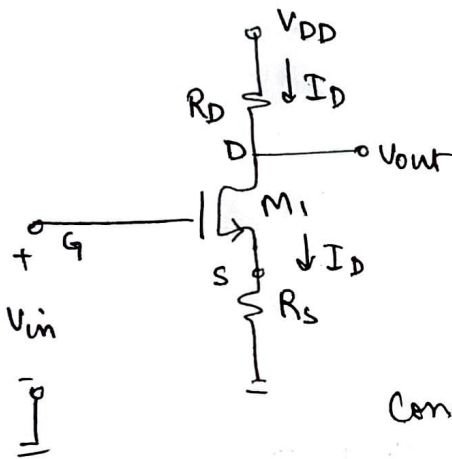
Summary :-

(11)

CS variations :



CS stage with degeneration:



Bias conditions :- [Assume $\lambda = 0$]

$$V_{in} = V_{GS} + I_D R_S$$

At saturation,

$$I_D = \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L}\right) (V_{GS} - V_{th})^2$$

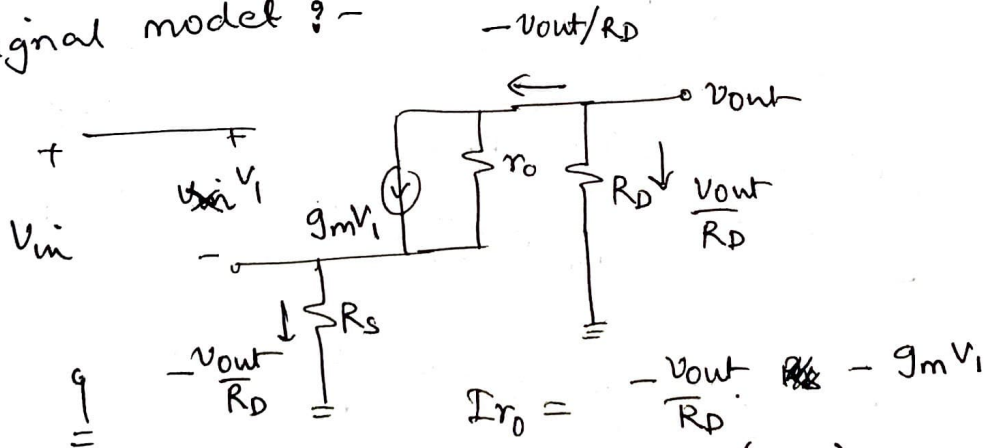
$$V_{out} = V_{DD} - I_D R_D$$

Condition that M_1 is in saturation:

$$V_{out} - V_{out} = V_{DD} - I_D R_D \geq V_{in} - V_{th}$$

Gain and I/O impedances:-

Small signal model :-



$$I_{r_o} = -\frac{V_{out}}{R_D} - g_m V_1$$

$$V_{in} = V_1 - \left(\frac{V_{out}}{R_D}\right) R_S$$

$$V_{out} = -\frac{V_{out}}{R_D} \cdot r_o - g_m V_1 r_o - \frac{V_{out}}{R_D} R_S$$

$$\text{or, } \left[1 + \frac{r_o}{R_D} + \frac{R_S}{R_D}\right] V_{out} = -g_m V_1 r_o \Rightarrow V_{out} = \frac{-g_m r_o V_1}{\left(1 + \frac{r_o}{R_D} + \frac{R_S}{R_D}\right)}$$

$$\therefore V_{out} = \frac{-g_m r_o}{1 + \frac{r_o}{R_D} + \frac{R_S}{R_D}} V_i$$

$$\therefore V_{in} = V_i - \frac{V_{out} R_S}{R_D}$$

$$= V_i - \frac{R_S}{R_D} \cdot \frac{(-g_m r_o) V_{in}}{\left[1 + \frac{r_o}{R_D} + \frac{R_S}{R_D}\right]}$$

$$\frac{V_{in}}{V_i} = 1 + \frac{g_m r_o R_S}{R_D + r_o + R_S}$$

$$\therefore V_{in} = \left[1 + \frac{g_m r_o R_S}{R_D + r_o + R_S}\right] V_i$$

$$\text{As } V_{out} = \frac{-g_m r_o}{1 + \frac{r_o}{R_D} + \frac{R_S}{R_D}} V_i$$

$$\text{and } V_i = V_{in} + \frac{V_{out} R_S}{R_D}$$

$$\therefore V_{out} = \frac{-g_m r_o}{\left[1 + \frac{r_o}{R_D} + \frac{R_S}{R_D}\right]} \left[V_{in} + V_{out} \frac{R_S}{R_D}\right]$$

$$\text{as, } V_{out} \left[1 + \frac{g_m r_o (R_S/R_D)}{\left[1 + \frac{r_o}{R_D} + \frac{R_S}{R_D}\right]}\right] = \frac{-g_m r_o}{\left[1 + \frac{r_o}{R_D} + \frac{R_S}{R_D}\right]} V_{in}$$

$$\text{as, } V_{out} \left\{1 + \frac{r_o}{R_D} + \frac{R_S}{R_D} + g_m r_o \frac{R_S}{R_D}\right\} = -g_m r_o V_{in}$$

$$\text{as, } \frac{V_{out}}{V_{in}} = \frac{-g_m r_o}{\left[1 + \frac{r_o}{R_D} + (1 + g_m r_o) \frac{R_S}{R_D}\right]}$$

$$A_v = \frac{-g_m r_o R_D}{R_D + r_o + (1 + g_m r_o) R_S}$$

$$= \frac{-R_D}{\left[\frac{R_D}{g_m r_o} + \frac{1}{g_m} + \left(\frac{1 + g_m r_o}{g_m r_o} \right) R_S \right]}$$

$$= \frac{-R_D}{\frac{1}{g_m} + R_S + \frac{R_S}{g_m r_o} + \frac{R_D}{g_m r_o}}$$

$$A_v = \frac{-R_D}{\frac{1}{g_m} + R_S + \frac{1}{g_m r_o} (R_S + R_D)}$$

if $R_S \rightarrow 0, r_o \rightarrow \infty,$

$$A_v \rightarrow \frac{-R_D}{(1/g_m)}$$

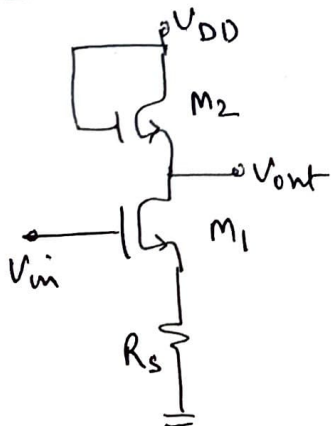
as $r_o \rightarrow \infty,$

$$A_v \rightarrow \frac{-R_D}{1/g_m + R_S}$$

The variation of $1/g_m$ w.r.t R_S is less, then A_v is almost constant.

$$= \frac{-\text{Resistance tied between drain \& ac ground.}}{\frac{1}{g_m} + \text{Resistance tied between source \& AC GND.}}$$

Example :-



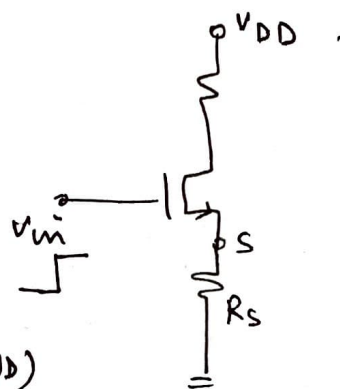
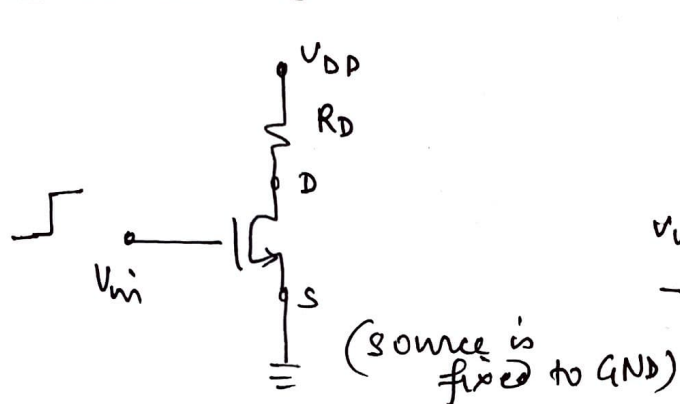
$$A_v = \frac{-g_{m2} (1/g_{m2})}{\frac{1}{g_{m1}} + R_S}$$

Note: if $\frac{1}{g_m} \ll R_S, A_v \rightarrow \frac{-R_D}{R_S}$

depends only on geometrical dimension.

Reason why gain is less in source degeneration :-

(14)



V_S

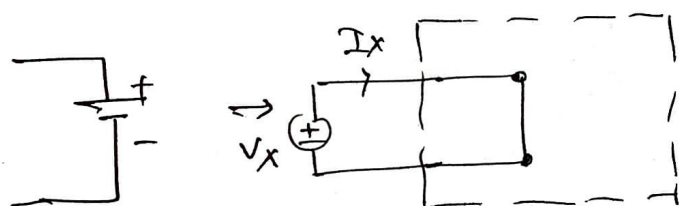
$\therefore I_D$ change less.
than $g_m \Delta V$.

$$V_{GS} \rightarrow V_{GS} + \Delta V$$

$$I_D \rightarrow I_{Ds} + g_m \Delta V$$

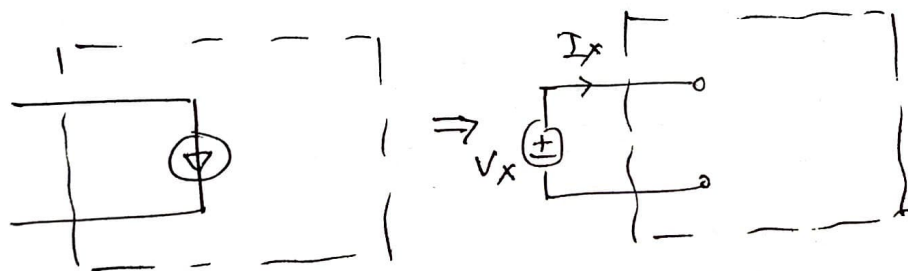
Impedances of ideal and constant current and voltage source:

case with a battery :-



$$R_{eq} = 0$$

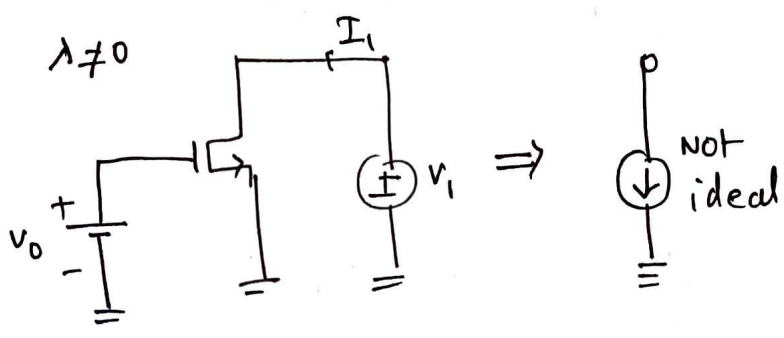
case with a current source :-



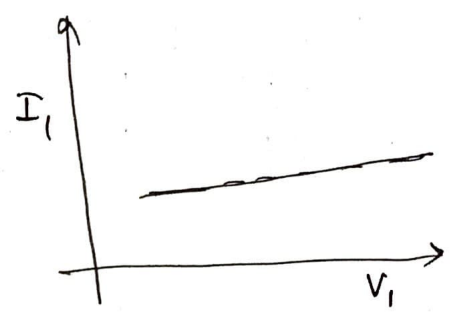
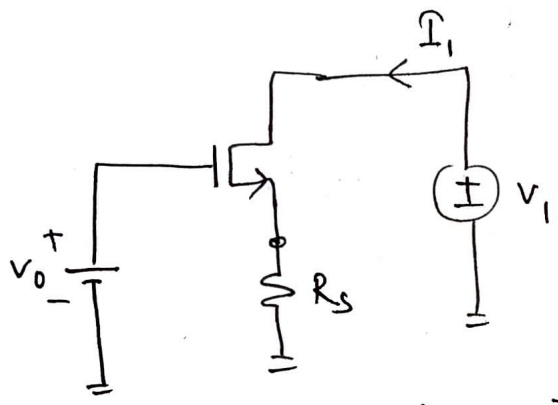
$$R_{eq} = \infty$$

let's build a better current source:

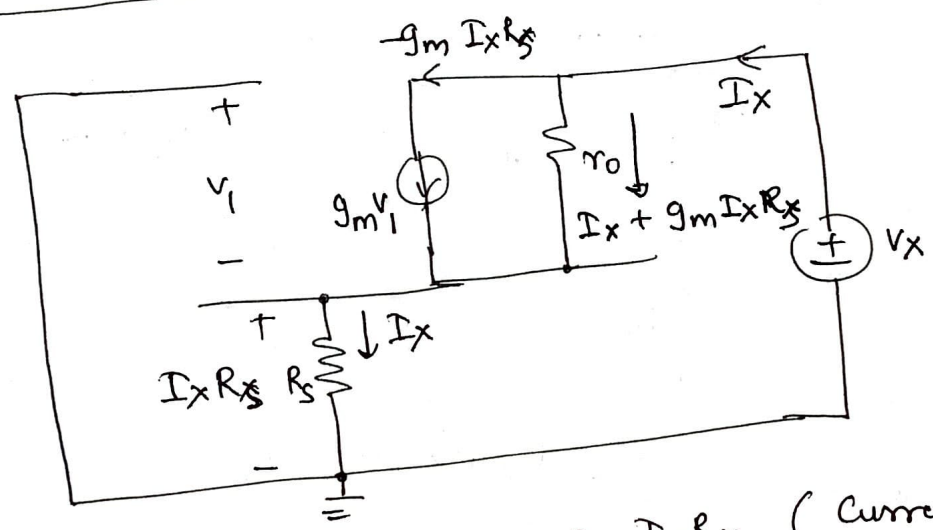
NOTE a better current source means at small signal,
 $R_{eq} \rightarrow \infty$.



A better current source :-



Look for small signal impedance :-



$V_1 = -I_x R_x$, $I_{x0} = I_x + g_m I_x R_x$ (Current from r_o).

$\therefore V_x = [I_x + g_m R_x I_x] r_o + I_x R_s$

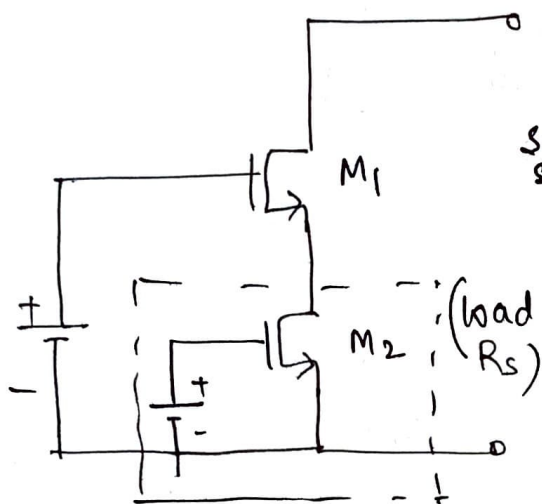
$\frac{V_x}{I_x} = (1 + g_m R_s) r_o + R_s$

$R_{out} = (1 + g_m R_s) r_o + R_s$

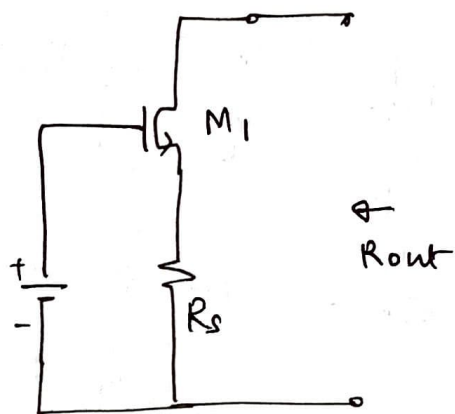
$R_{out} = \underbrace{(1 + g_m r_o) R_s + r_o}_{\text{(Net increased)}} \uparrow \text{this was when } R_s \rightarrow 0.$

Example:- Replace R_S with current source.

(16)



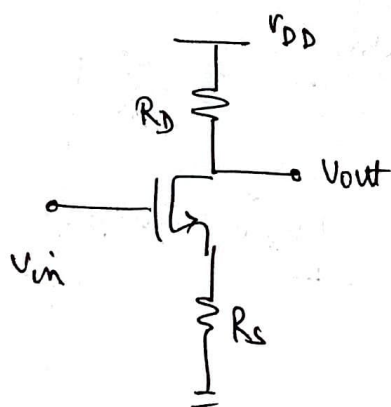
Small signal
→



$$\therefore R_{out} = (1 + g_{m1} r_{o1}) r_{o2} + r_{o1}$$

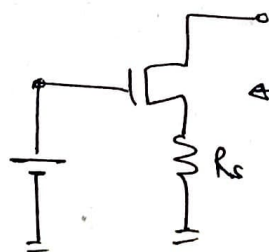
Biasing techniques and common gate.

Review:-



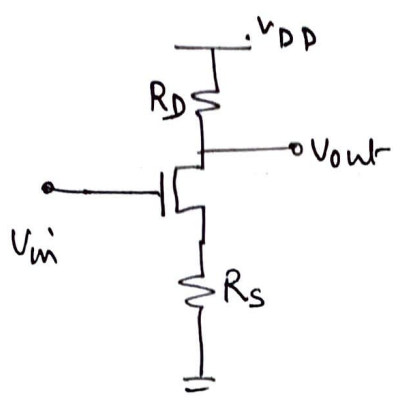
$$A_v = \frac{-R_D}{1/g_m + R_S}, \quad \lambda = 0.$$

Current source with degeneration:-



$$R_{eq} = (1 + g_m r_o) R_S + r_o.$$

I/O Impedance in degenerate CS stage-

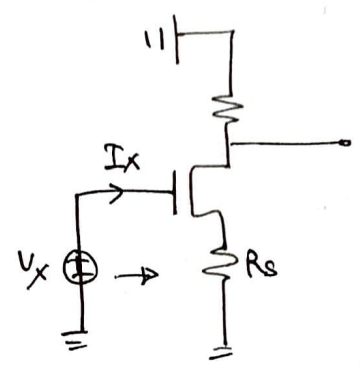


Small signal model:-

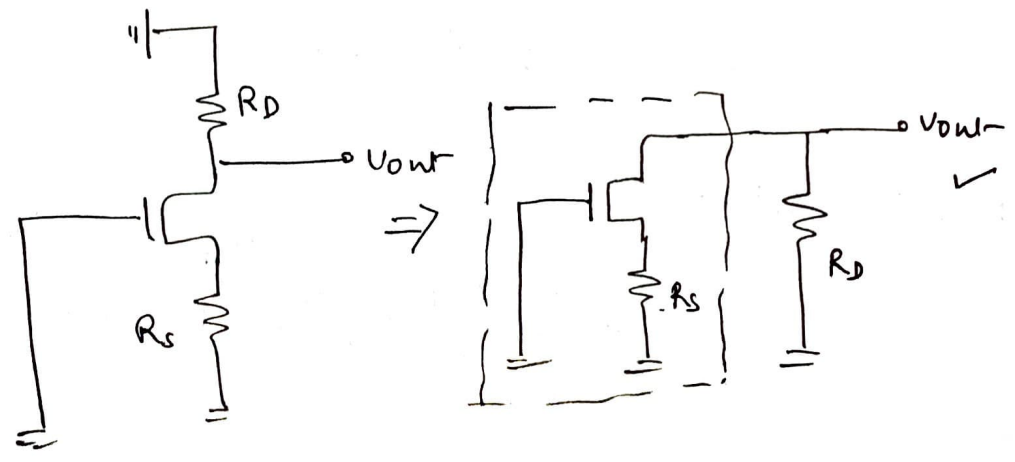
Input impedance :

$$I_x = 0$$

$$\Rightarrow R_{in} = \infty.$$



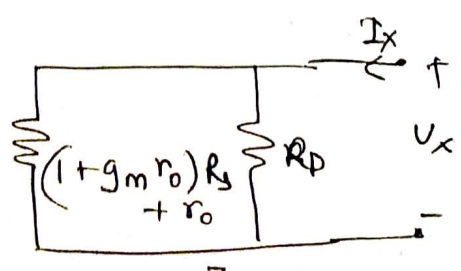
Output impedance at $\lambda \neq 0$,



Now,

$$R_{out} = (1 + g_m r_o) R_S + r_o$$

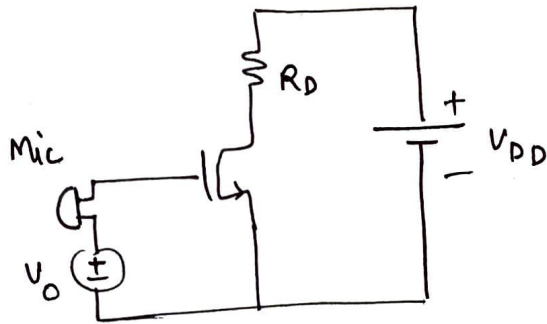
$$\therefore V_{out} =$$



$$\therefore R_{out} = R_D \parallel [(1 + g_m r_o) R_S + r_o]$$

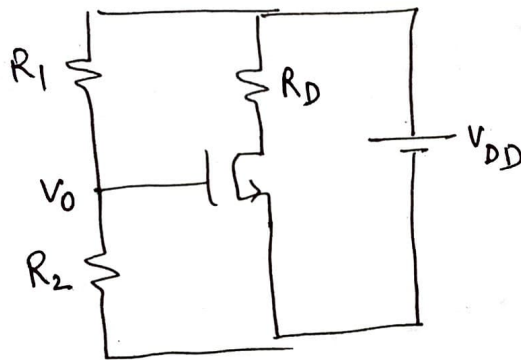
Biasing technique of CS-stage:-

(18)



Too many batteries!

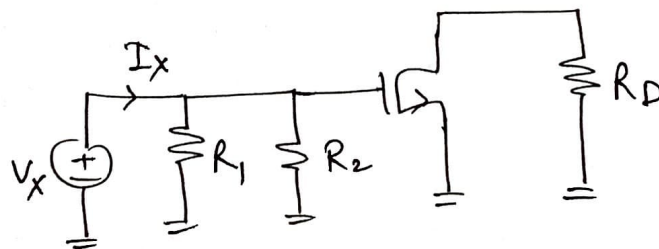
use a resistive divider



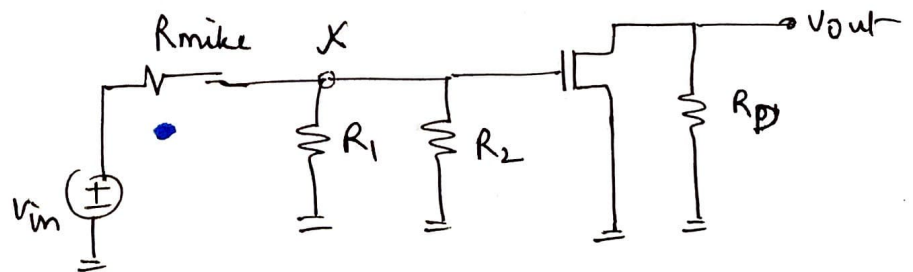
$$V_0 = \frac{R_2}{R_1 + R_2} V_{DD}$$

Now, we have (for $\lambda = 0$),

$$\underline{\underline{R_{in}}} = R_1 \parallel R_2$$



AC voltage gain:-



$$\therefore \frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_x} \cdot \frac{V_x}{V_{in}}$$

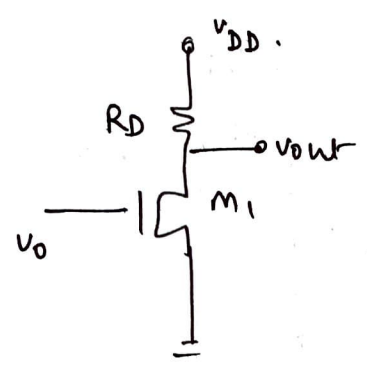
$$\text{Now, } \frac{V_x}{V_{in}} = \frac{R_1 \parallel R_2}{R_1 \parallel R_2 + R_{mic}}$$

$$\frac{V_{out}}{V_x} = -g_m R_D$$

$$\therefore A_v = \frac{R_1 \parallel R_2}{[R_1 \parallel R_2 + R_{mic}]} \cdot (-g_m R_D)$$

To increase gain, choose $R_1 \parallel R_2 \rightarrow \infty$
 $R_1 \parallel R_2 \gg R_{mic}$

Can we increase R_D to increase gain?



for M_1 to be in saturation

$$V_{out} \geq V_0 - V_{th}$$

$$\therefore V_{DD} - I_D R_D \geq V_0 - V_{th}$$

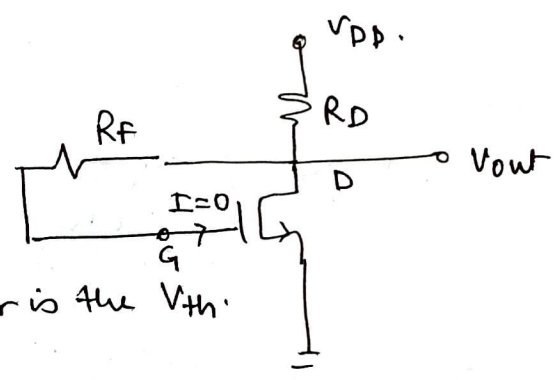
$$\therefore \frac{V_{DD} + V_{th} - V_0}{I_D} \geq R_D$$

$$\text{or, } \frac{V_{DD} - V_0 + V_{th}}{I_D} \geq R_D$$

$$\text{or, } R_D \leq \frac{V_{DD} - V_0 + V_{th}}{I_D} \text{ for } M_1 \text{ in saturation}$$

Self correction biasing :-

Say I take two wafers and design the amplifier. But, only thing that got changed in these two wafers is the V_{th} . Will R_F help in that case?

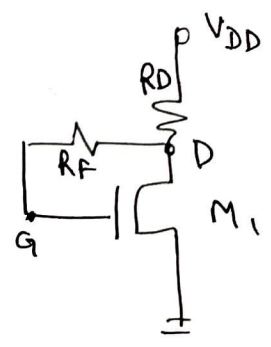


$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{gs} - V_{th})^2$$

Now once $V_{th} \uparrow$, $I_D \downarrow$ or $V_{th} \downarrow$, $I_D \uparrow$.

If $I_D \uparrow$ then $V_{out} \downarrow$. But $V_{DS} = V_{GS}$ as $I_G = 0$

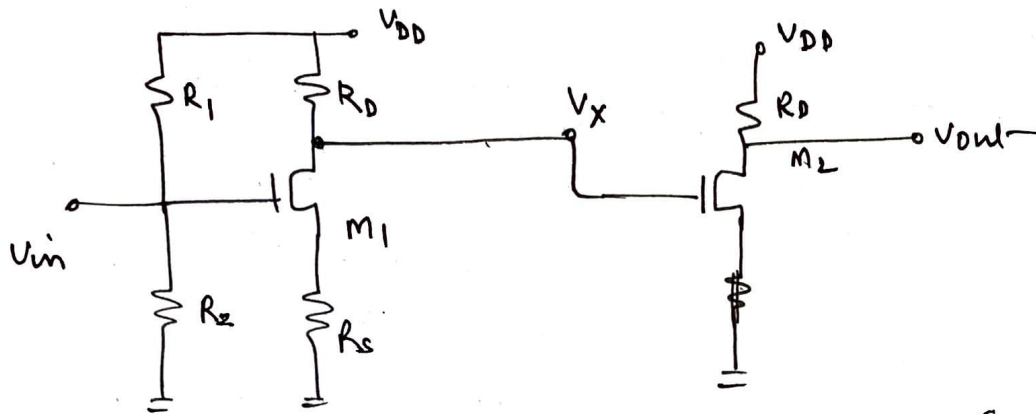
Thus, if $V_{DS} \downarrow$ then V_{GS} also $\downarrow$ thus maintaining I_D .



The M_1 is always in saturation as G and D are connected to same potential.

Example of cascade

(20)

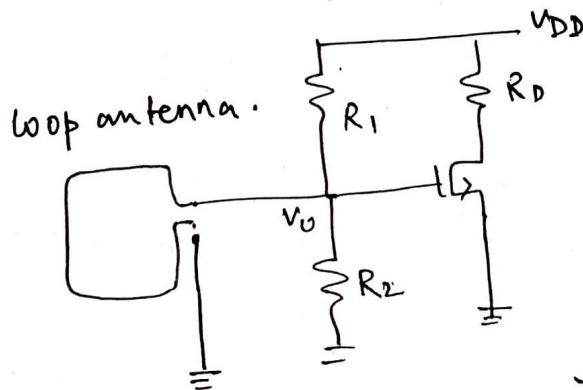


$$\frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_x} \cdot \frac{V_x}{V_{in}}, \quad \frac{V_x}{V_{in}} = - \frac{R_D}{\frac{1}{g_{m1}} + R_E}$$

$$\frac{V_{out}}{V_x} = -g_{m2} R_D$$

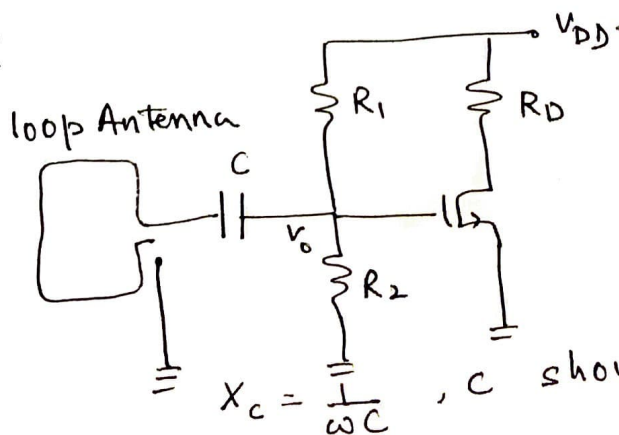
$$\therefore A_v = +g_{m2} R_D \cdot \left(\frac{R_D}{\frac{1}{g_{m1}} + R_E} \right)$$

Biasing with an antenna:-



Note here that if there is no signal, the antenna is just like a short-circuit. $\therefore V_0 = 0$ i.e. the transistor is ~~short~~ off.

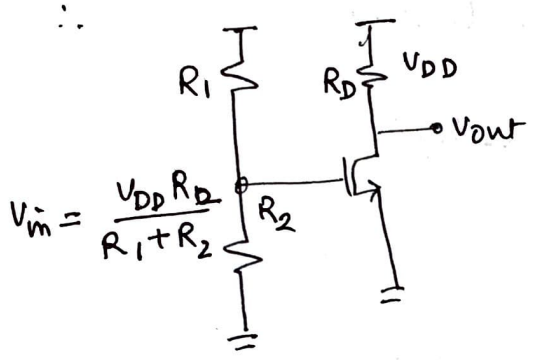
How to fix?



$X_C = \frac{1}{\omega C}$, C should be large enough.

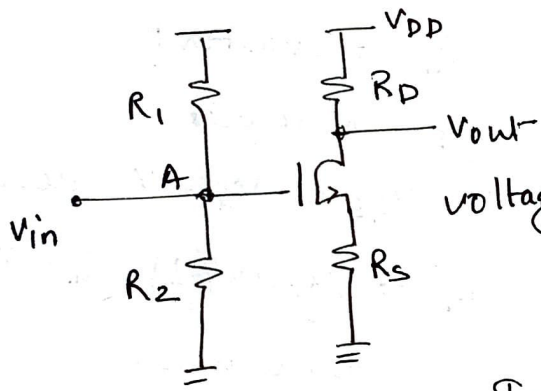
$$I_D = \frac{\mu_n}{2} C_{ox} \left(\frac{W}{L} \right) \left[\frac{V_{DD} R_D}{R_1 + R_2} - V_{th} \right]^2$$

So, see I_D is strongly dependent on V_{DD} , V_{th} , $\mu_n C_{ox}$, temper.



This circuit is thus not a much stable circuit.

∴ To reduce the sensitivity, use degenerate CS stage:-



This circuit is less sensitive to the voltage change at the node A.

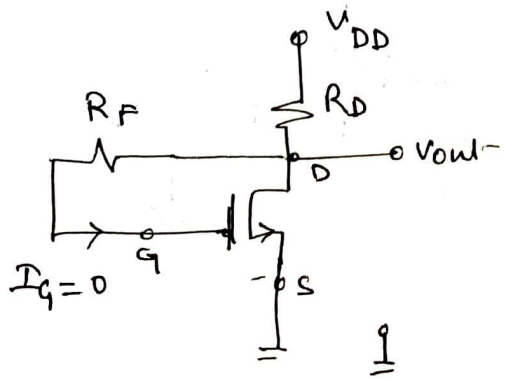
$$V_{in} = \frac{V_{DD} R_2}{R_1 + R_2}$$

$$I_D = \frac{\mu_n}{2} C_{ox} \left(\frac{W}{L} \right) \left[\frac{V_{DD} R_2}{R_1 + R_2} - V_{th} \right]^2 \rightarrow V_{GS} \text{ now.}$$

Hence, $V_{in} = \frac{V_{DD} R_2}{R_1 + R_2} = V_{GS} + I_D R_S$

$$I_D = \frac{\mu_n}{2} C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_{th})^2$$

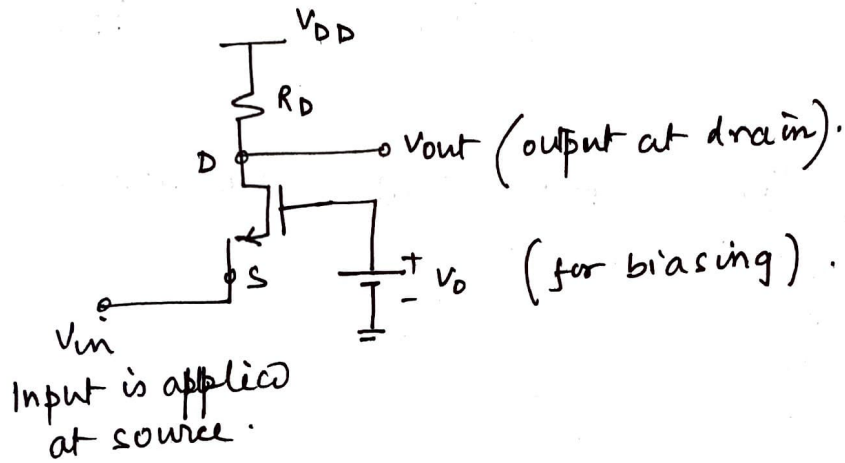
~~Thus~~, A much better CS stage to reduce the sensitivity is:-



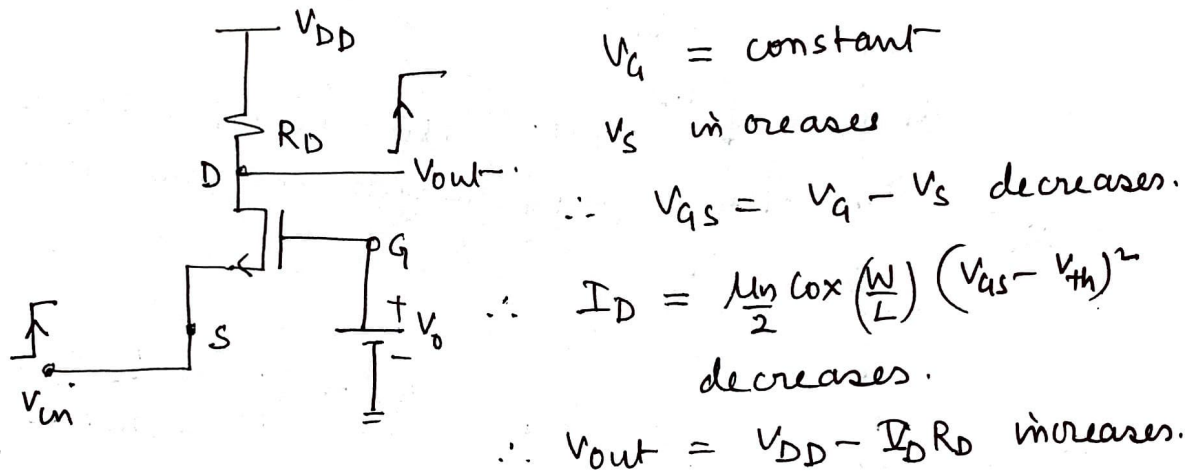
$V_{DS} = V_{GS}$ as $I_D \Rightarrow 0$
 this ckt is always at saturation
 ∴ $V_D = V_G$

Common Gate variations.

(22)

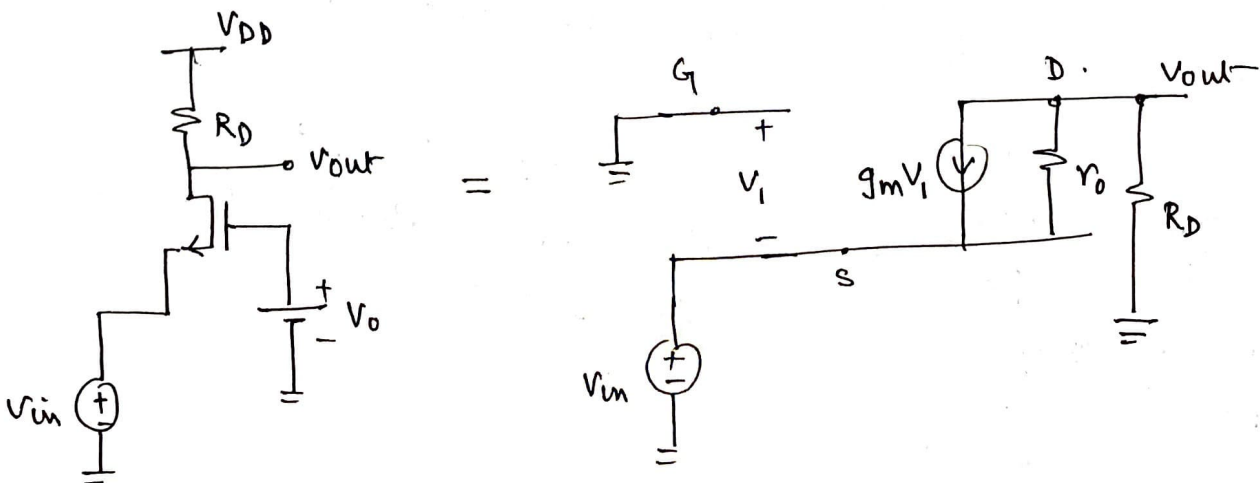


let's say now the input is changed by ΔV



$\therefore$ the gain does not get reversed.
 $\therefore$ gain is positive.

Small-signal analysis: $(\lambda = 0)$



Here $V_{in} = -V_i$

$$\therefore g_m V_i = -g_m V_{in}$$

for $r_o \rightarrow \infty$,

$$\therefore \frac{V_{out}}{R_D} - g_m V_{in} = 0$$

$$\therefore \boxed{\frac{V_{out}}{V_{in}} = +g_m R_D}$$

Input resistance:-

$$\therefore I_x = -g_m V_i$$

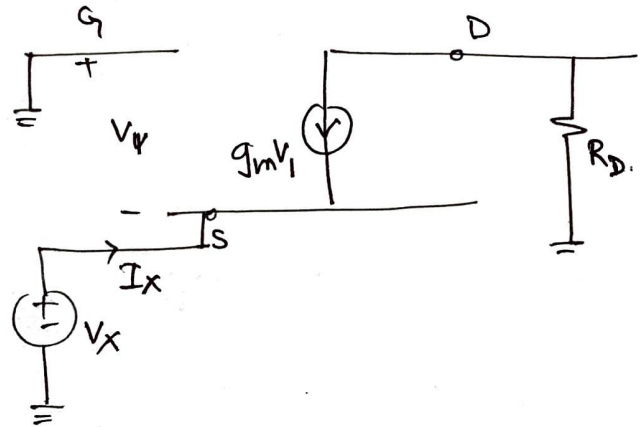
$$\& V_i = -V_x$$

$$\therefore I_x = g_m V_x$$

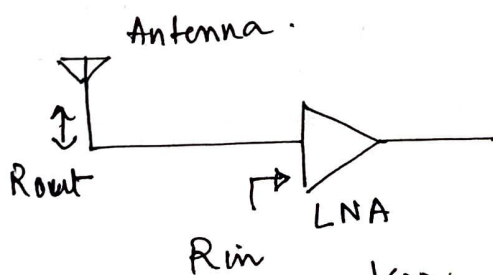
$$\therefore \frac{V_x}{I_x} = \frac{1}{g_m}$$

$$\therefore \boxed{R_{in} = \frac{1}{g_m}}$$

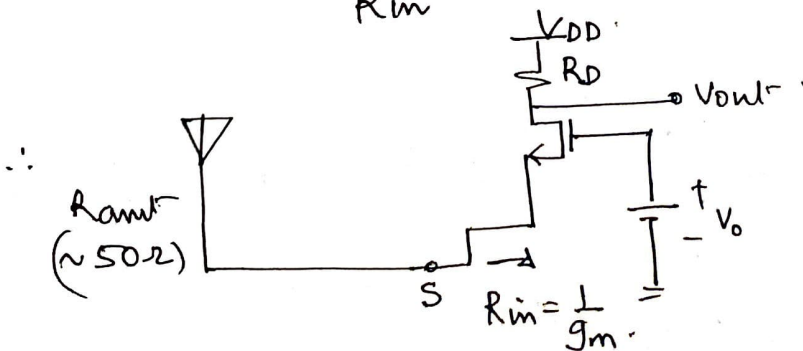
This is not infinite as seen in CS stage.



Application:- Bluetooth receiving antenna:-

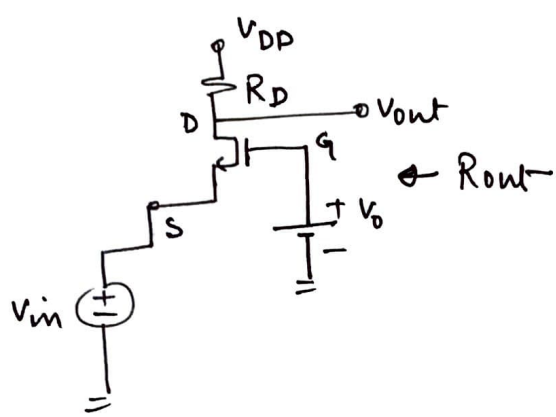


Requirement is R_{in} of LNA
 $= R_{out} \cdot (\sim 50 \Omega)$.

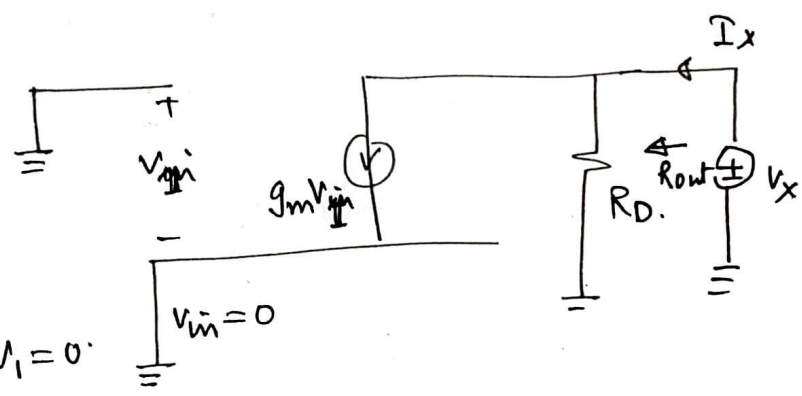


$$\text{set } g_m = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right) I_D} = 50 \Omega$$

Output Impedance :-

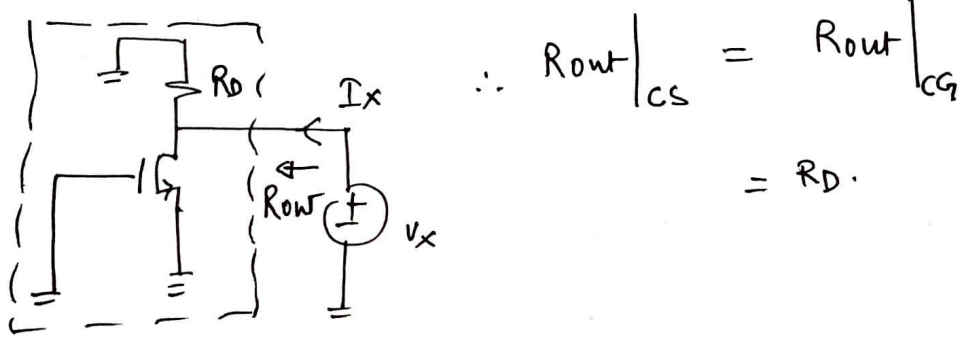


Small signal analysis.



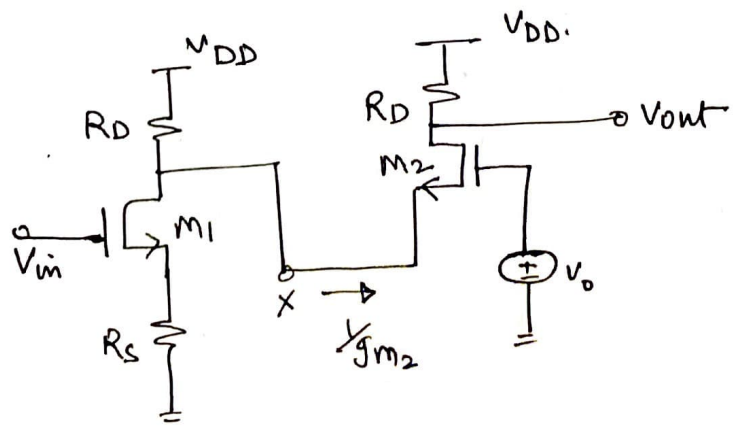
$$R_{out} = \frac{V_x}{I_x}, V_i = 0$$

compare this with the CS-stage.



$$\therefore R_{out}|_{CS} = R_{out}|_{CG} = R_D$$

Example :-



what is $\frac{V_{out}}{V_{in}}$?
($\lambda = 0$)

$$\frac{v_{out}}{v_{in}} = \frac{v_{out}}{v_x} \cdot \frac{v_x}{v_{in}}$$

Now,

$$\frac{v_x}{v_{in}} = \frac{-g_m(R_D \parallel \frac{1}{g_{m2}})}{\frac{1}{g_{m1}} + R_S}$$

Note :-

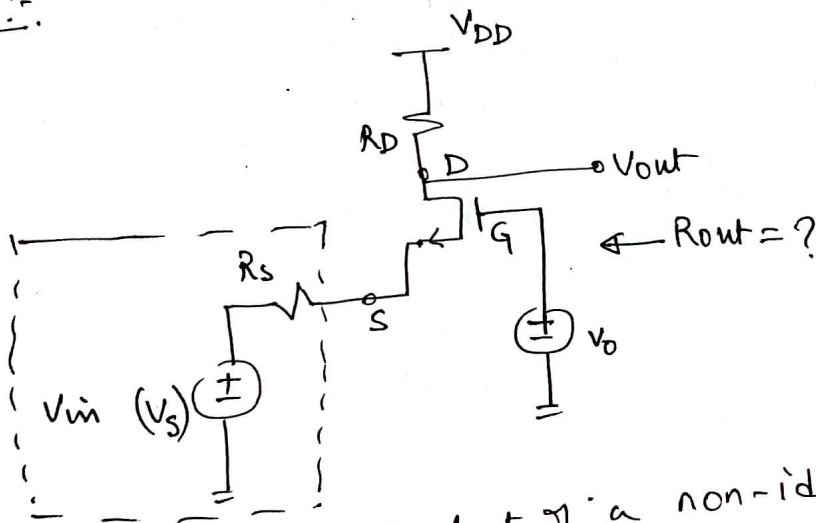
$\frac{1}{g_{m2}}$ is the resistance tied between D & AC GND. This is R_{in} of M_2 .

$$\frac{v_{out}}{v_x} = g_{m2} R_D$$

$$\therefore \frac{v_{out}}{v_{in}} = \frac{-(R_D \parallel \frac{1}{g_{m2}})}{\frac{1}{g_{m1}} + R_S} \times g_{m2} R_D$$

$$A_v = \frac{-(R_D \parallel \frac{1}{g_{m2}})}{\frac{1}{g_{m1}} + R_S} \times g_{m2} R_D$$

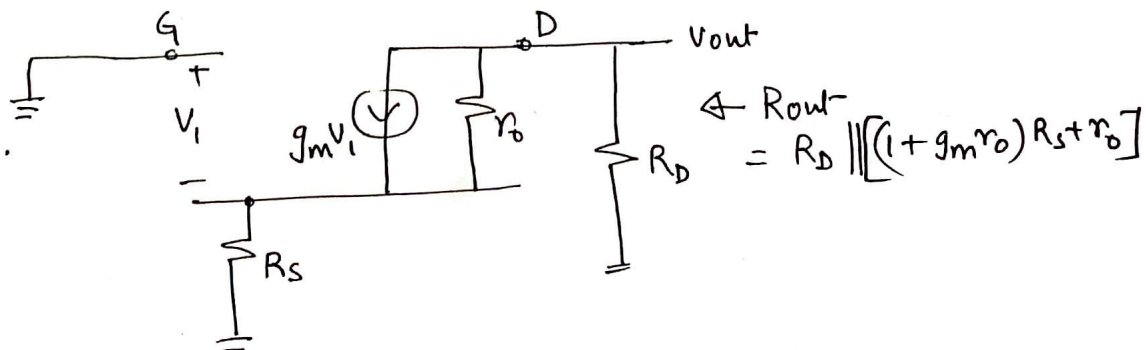
Output Impedance of CG stage with non-ideal voltage source at input:-



$(\lambda \neq 0)$

Thevenin's equivalent of a non-ideal voltage source.

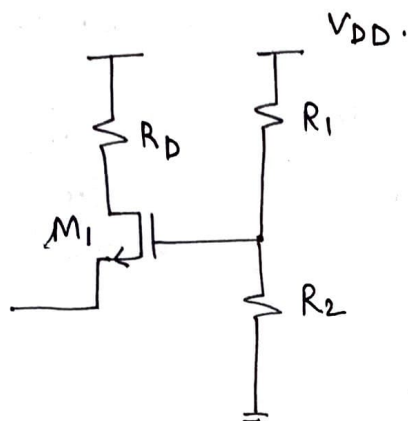
This is a CG stage with degeneration. !!!



$$R_{out} = R_D \parallel [(1 + g_m r_o) R_S + r_o]$$

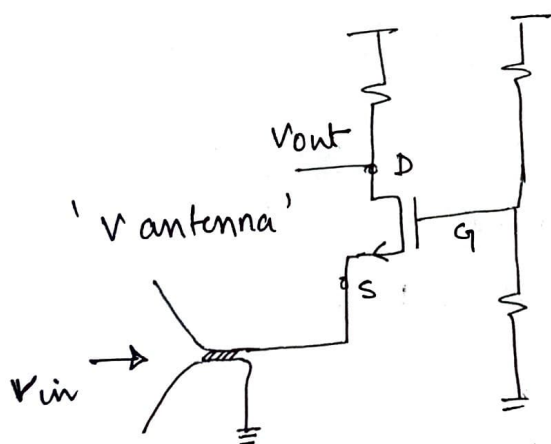
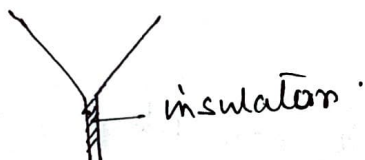
Bias design in CG state:

(26)



Now there is a problem with the type of say antenna at the input :-

If the antenna is 'V' shaped.

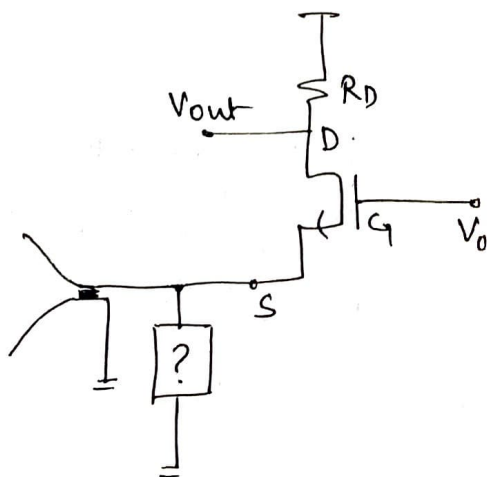


If there is no signal, the antenna is open, thus source is disconnected for d.c.

$$\therefore I_D = 0$$

$$\therefore g_m = 0$$

Thus. in order that $I_D \neq 0$ (saturation), one has to put a 2-terminal device.

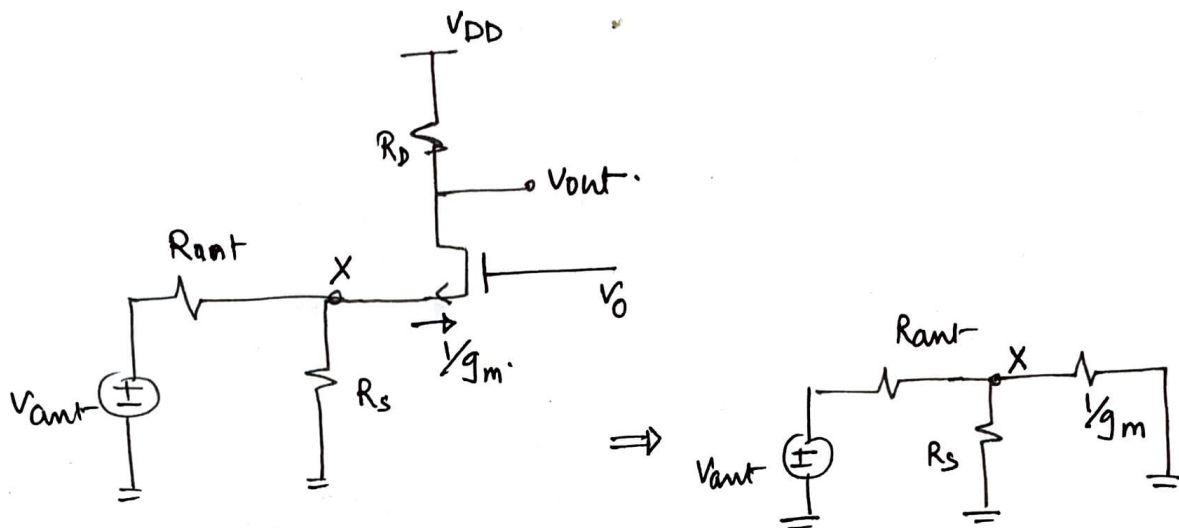


$$\boxed{?} = R, I_D, L$$

Voltage gain ($\lambda = 0$),

(28)

$$\frac{V_{out}}{V_{ant}} = ?$$



$$\frac{V_x}{V_{ant}} = \frac{R_s \parallel \frac{1}{g_m}}{\left(\frac{1}{g_m} \parallel R_s\right) + R_{ant}} = \frac{R_s \parallel \frac{1}{g_m}}{(R_s \parallel \frac{1}{g_m}) + R_{ant}}$$

$$\therefore \frac{V_{out}}{V_{ant}} = \frac{V_{out}}{V_x} \cdot \frac{V_x}{V_{ant}}, \quad \frac{V_{out}}{V_x} = g_m R_D.$$

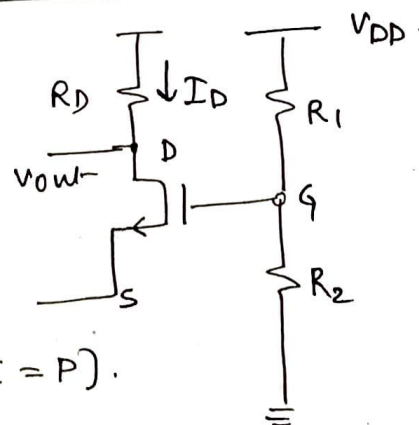
$$= \frac{R_s \parallel \frac{1}{g_m}}{R_s \parallel \frac{1}{g_m} + R_{ant}} \times g_m R_D.$$

Example: $V_{DD} = 1.5V$, $P = 1.5mW$

$$\mu n C_{ox} = 100 \mu A/V^2$$

$$V_{th} = 0.5V, \lambda = 0$$

$$W/L = 50.$$



Max Power limit = $1.5mW$

$$\therefore \text{Max Current} = 1mA \quad (VI = P).$$

$$\text{i.e., } I_{R_1} \sim 0, \quad I_D \sim 1mA.$$

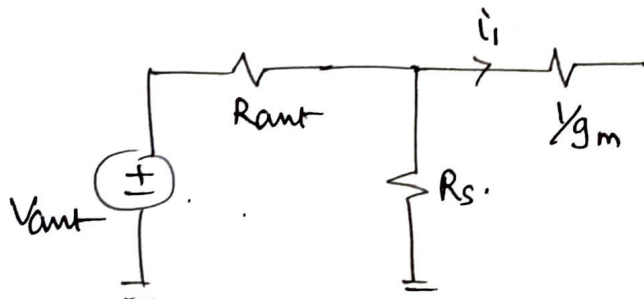
$$I_D = \frac{1}{2} \mu n C_{ox} \left(\frac{W}{L}\right) \cdot (V_{GS} - V_{th})^2.$$

$$\therefore V_{GS} = 0.63V, \quad V_{GS} - V_{th} = 0.63V, \quad V_{GS} = 1.13V$$

$$g_m = \frac{2 I_D}{V_{GS} - V_{th}} \quad (\text{Take any of the 3 eqns. of } g_m)$$

$$= \frac{3.2 \text{ mA}}{316 \Omega} = \frac{1}{316 \Omega}$$

thus, for this antenna case,



$$i_1 = \frac{R_s}{R_s + 1/g_m} i_{ant}$$

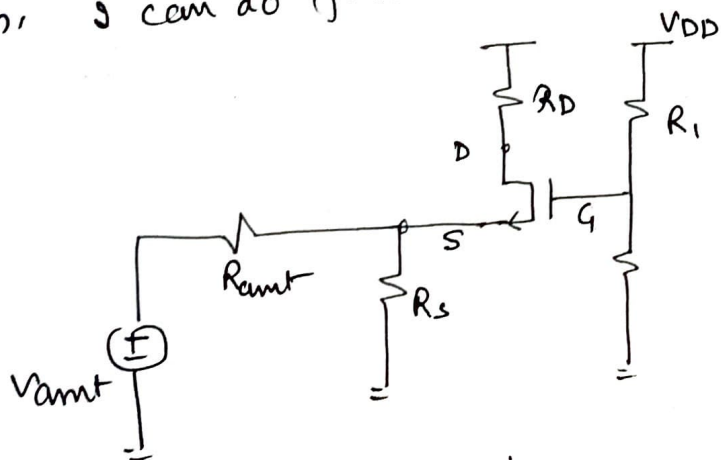
$$\text{2 } R_s \gg 1/g_m$$

$$\text{choose } R_s = 10 \left(\frac{1}{g_m} \right) = 3.16 \text{ k}\Omega$$

Now, if you choose this $3.16 \text{ k}\Omega$, then, voltage across R_s is

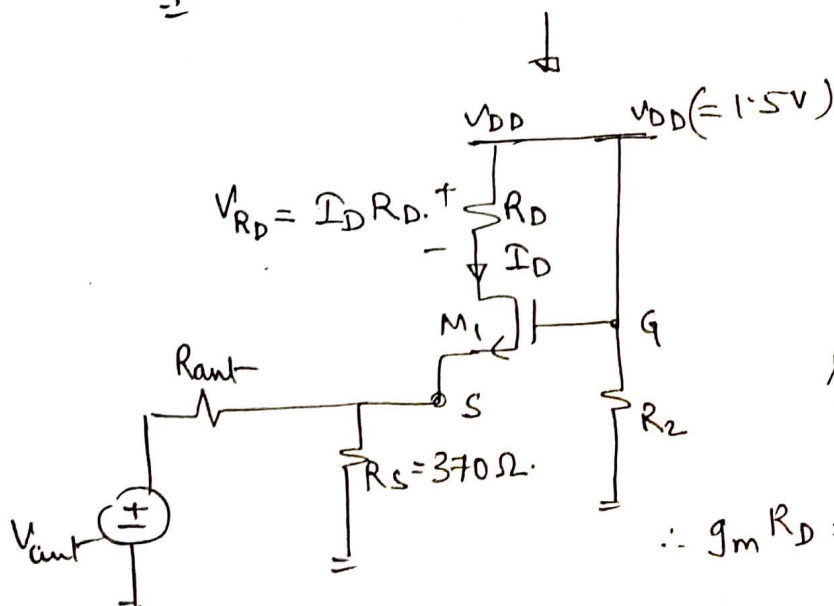
$$I_D \cdot R_s = 1 \text{ mA} \times 3.16 \text{ k}\Omega = 3.16 \text{ V} > V_{DD} !!!$$

thus, I can do find max voltage available at R_s :-
Make R_1 as low as possible



Then!
 $R_s \Big|_{\text{max}}$

$$\begin{aligned} R_s &= \frac{V_{DD} - V_{GS}}{1 \text{ mA}} \\ &\approx \frac{V_{DD} - 0.63}{1 \text{ mA}} \\ &\approx 370 \Omega \end{aligned}$$



for M_1 to be in saturation,

$$I_D R_D \leq V_{th}$$

$$\therefore R_D = 500 \Omega$$

If $R_D > 500 \Omega$, $V_{RD} \rightarrow$ goes to $M_1 \rightarrow$ triode region

$$\therefore g_m R_D = \frac{500}{316} \approx 1.6$$

Say now, you don't want to increase Power. i.e. power is limited. $\therefore I_D$ is limited.

How you can increase gain?

$$g_m = \sqrt{2\mu_n C_{ox} \cdot \left(\frac{W}{L}\right) I_D}$$

You can play by increasing $\frac{W}{L}$ twice. its value.

$$\left.\frac{W}{L}\right|_{\text{old}} = 50, \quad \left.\frac{W}{L}\right|_{\text{new}} = 100.$$

So g_m will increase, but the device will be slower as larger device has large capacitances.

$$\therefore I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_{\text{new}} (V_{GS} - V_{th})^2.$$

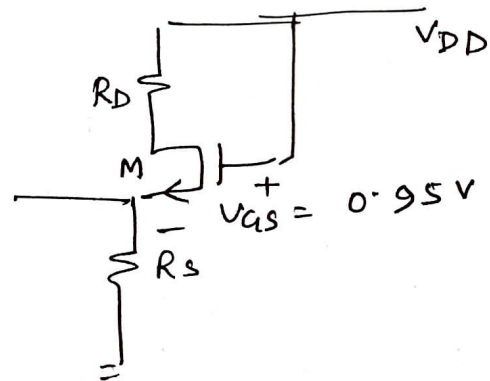
$$I_D = 1 \text{ mA}.$$

$$\therefore V_{GS} = 0.95 \text{ V}.$$

$$\therefore R_S|_{\text{new}} = \frac{V_{DD} - V_{GS}}{I_D} = 550 \Omega.$$

$$\therefore g_m|_{\text{new}} = 4.5 \text{ mS} = \frac{1}{220 \Omega}$$

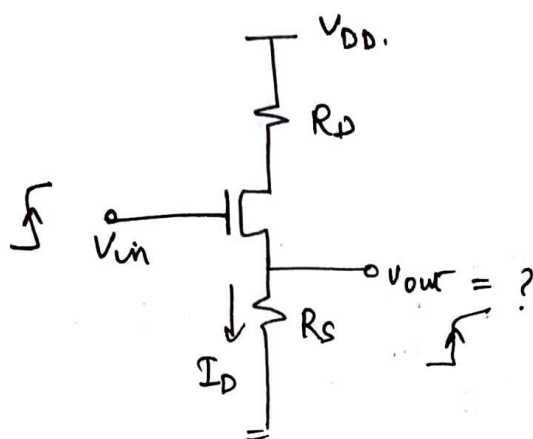
$$\therefore g_m R_D = 2.3$$



New g_m is less $\therefore$ more current into M.

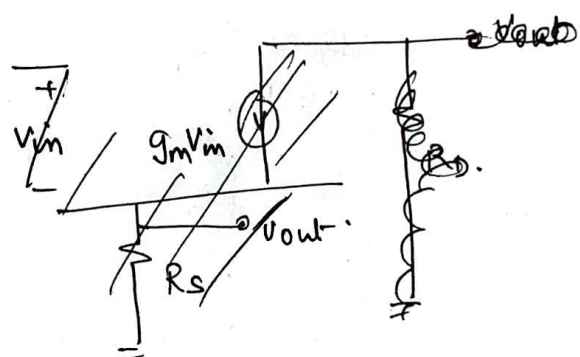
let's apply a step ΔV at V_{in}

(32)



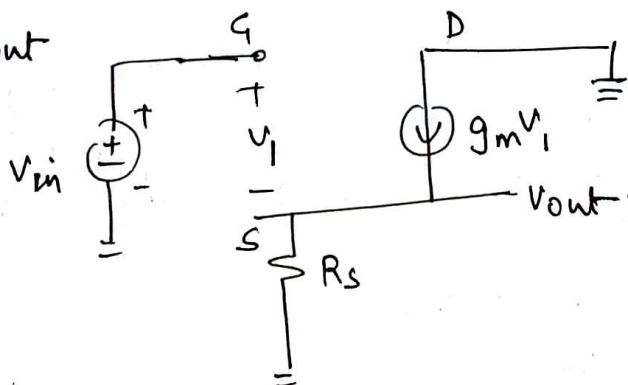
$V_{in} (\uparrow)$, $V_{out} = ?$
 if $V_{in} (\uparrow)$, and if V_{out} does not increase,
 then $V_{GS} = V_G - V_S$ increases
 $\therefore I_D$ increases.

Small signal model:-



$$V_{in} = V_1 + V_{out}$$

$$\therefore V_1 = V_{in} - V_{out}$$



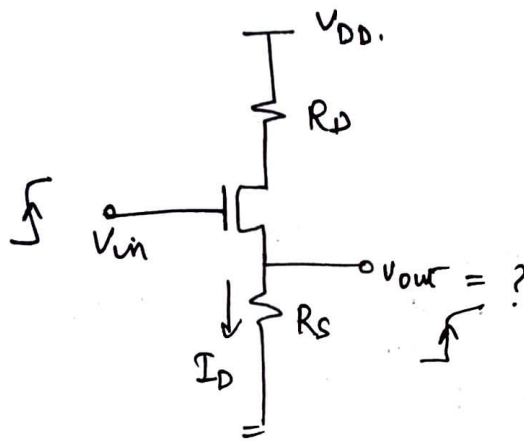
$$\therefore g_m (V_{in} - V_{out}) = \frac{V_{out}}{R_S}$$

$$\therefore \frac{V_{out}}{V_{in}} = \frac{R_S}{\frac{1}{g_m} + R_S}$$

$$A_v = \frac{\text{Total resistance tied between S and AC GND.}}{\frac{1}{g_m} + \text{Total resistance tied between S and AC GND.}}$$

let's apply a step Δv at v_{in}

(32)



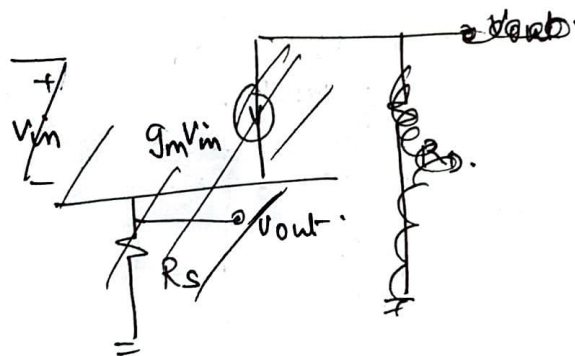
$v_{in}(\uparrow)$, $v_{out} = ?$

if $v_{in}(\uparrow)$, and if v_{out} does not increase,

then $V_{GS} = V_G - V_S$ increases

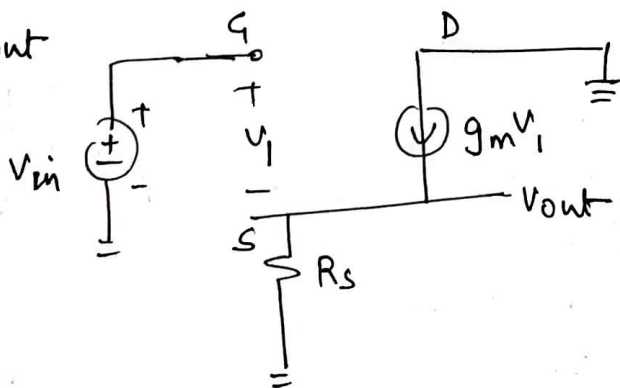
$\therefore I_D$ increases.

Small signal model:-



$$v_{in} = v_1 + v_{out}$$

$$\therefore v_1 = v_{in} - v_{out}$$

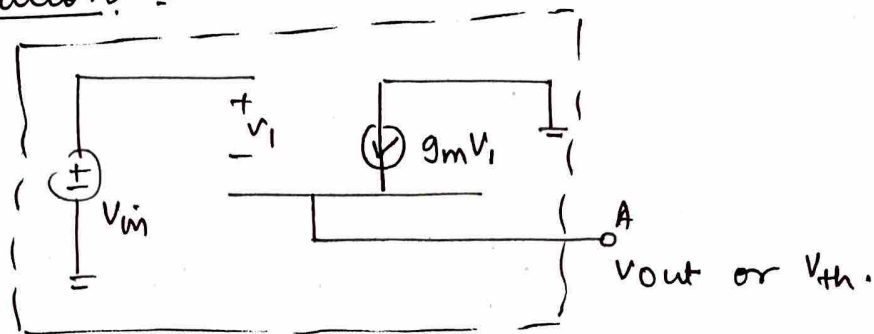


$$\therefore g_m (v_{in} - v_{out}) = \frac{v_{out}}{R_S}$$

$$\therefore \frac{v_{out}}{v_{in}} = \frac{R_S}{\frac{1}{g_m} + R_S}$$

$$A_v = \frac{\text{Total resistance tied between S and AC GND.}}{\frac{1}{g_m} + \text{Total resistance tied between S and AC GND.}}$$

Alternative derivation :-

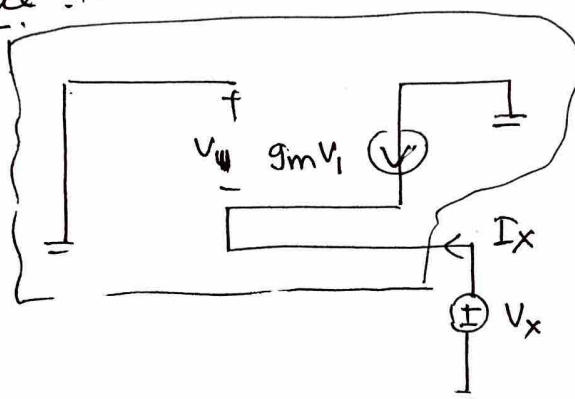


Thevenin's voltage :-

Here, V_{th} port A is open ckt. $\therefore g_m V_1 = 0 \Rightarrow V_1 = 0$.

$$\therefore V_{out} = V_{th} = V_{in}.$$

Thevenin's resistance :-

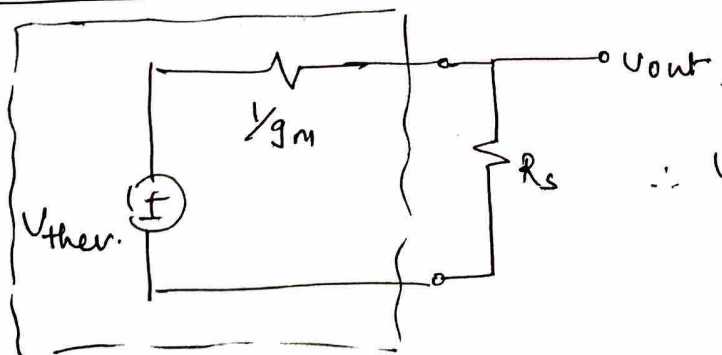


$$V_1 = V_x$$

$$\therefore \frac{V_x}{I_x} = \frac{1}{g_m}.$$

This is equivalent to diode connected source stage.

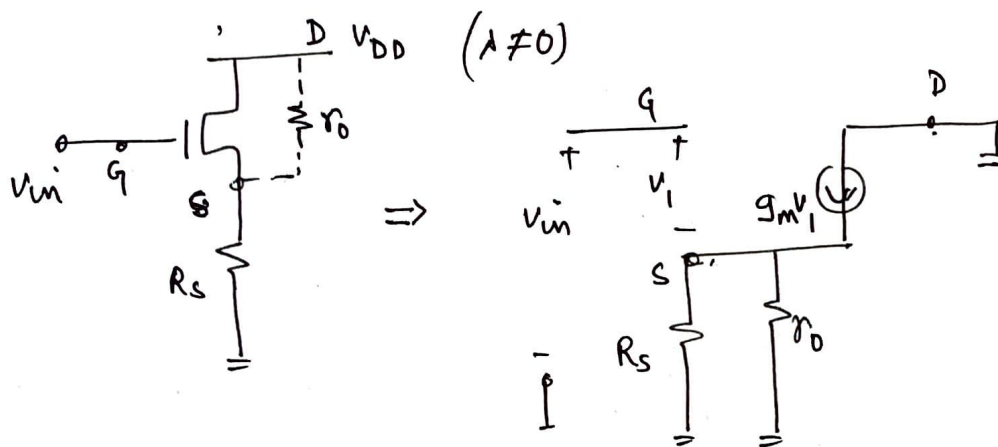
SS model :-



$$\therefore V_{out} = \frac{R_s \cdot V_{th}}{R_s + 1/g_m}.$$

In presence of channel length modulation :-

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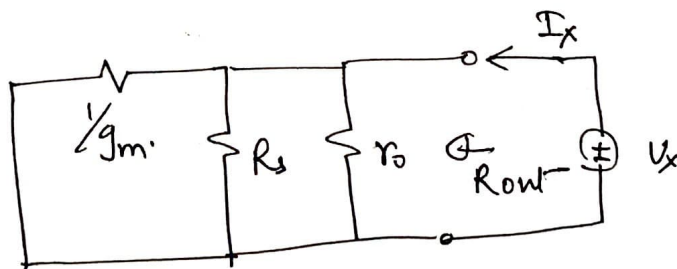
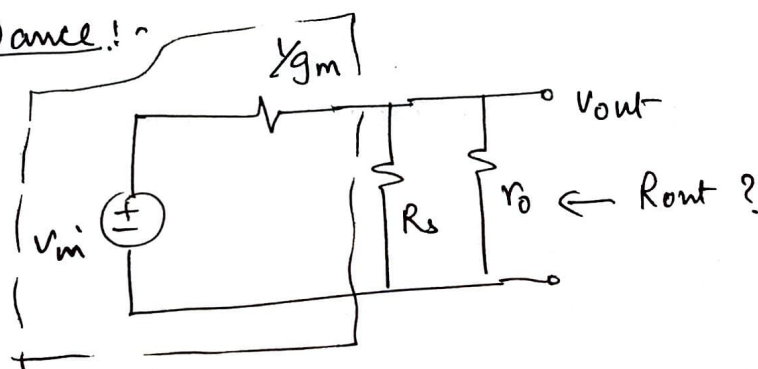


Note r_o is between D & S
But Drain is ground in ac ss analysis.

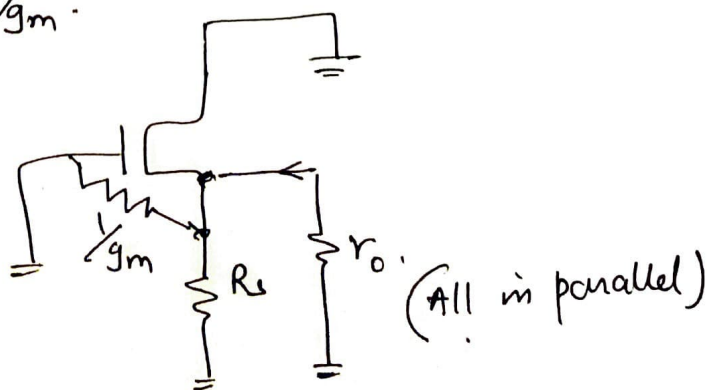
$$\therefore \frac{V_{out}}{V_{in}} = \frac{R_s \parallel r_o}{\frac{1}{g_m} + R_s \parallel r_o}$$

$$R_{in} = \infty.$$

Output Impedance :-

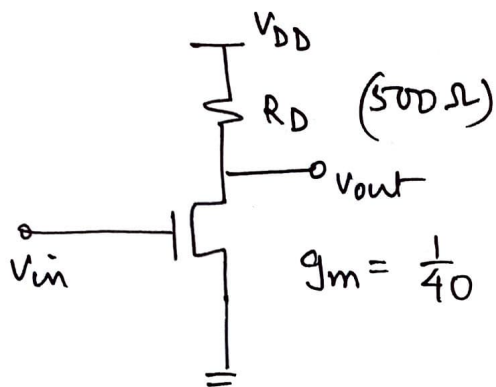


$$R_{out} = r_o \parallel R_s \parallel \frac{1}{g_m}$$



Application of source follower.

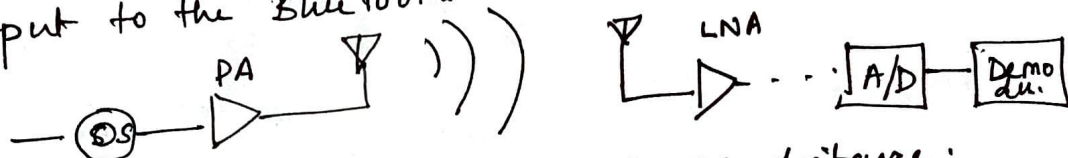
35



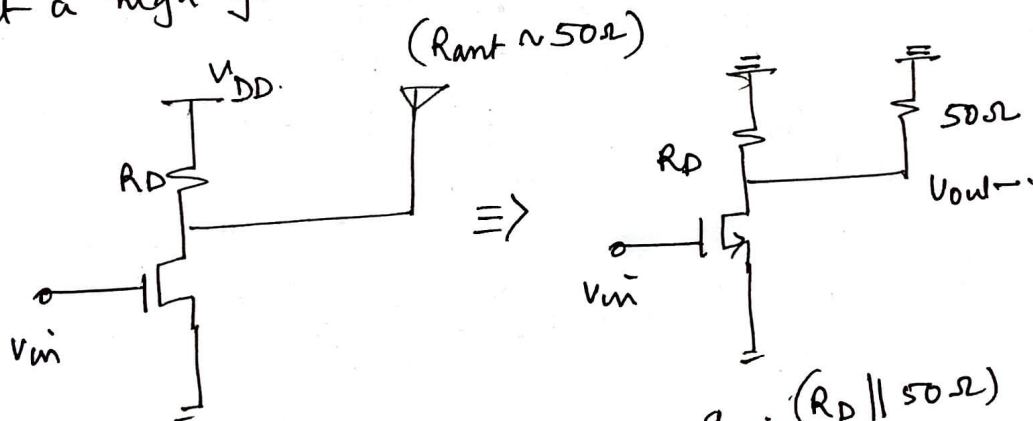
$$A_v = -g_m R_D$$

$$= -\frac{500}{40} = -12.5$$

Let's put this output to the Bluetooth transmitter antenna:-



We want a high gain so that it can cover large distance.



$$\therefore \frac{v_{out}}{v_{in}} = -g_m (R_D \parallel 50\Omega)$$

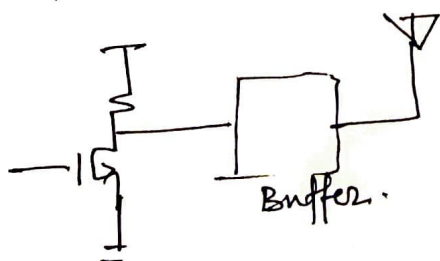
$$= -\frac{1}{40} \times (500 \parallel 50)$$

$$= 1.14 !!!$$

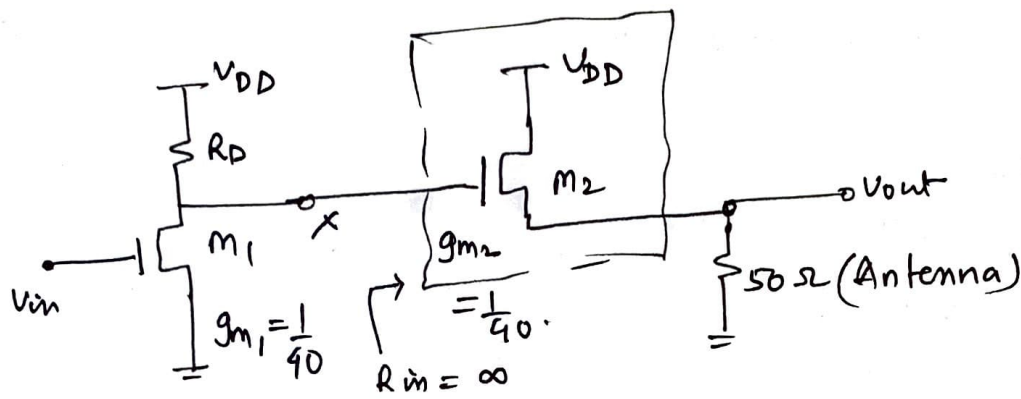
$\therefore$ Gain reduced.

Note that load is heavy $\rightarrow$ means it draws lots of current.

$\therefore$ We need to put in between something.



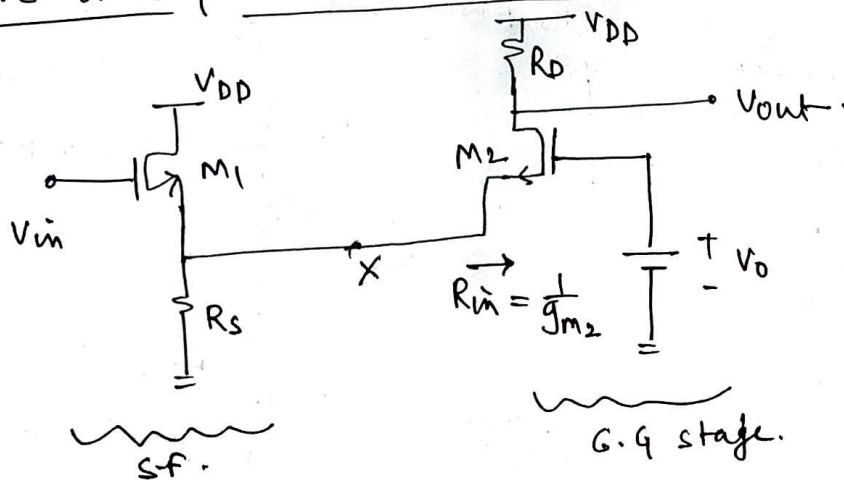
Buffer tries to drive the heavy load by keeping the gain intact.



Now:-

$$\begin{aligned} \frac{V_{out}}{V_{in}} &= \frac{V_{out}}{V_x} \times \left(\frac{V_x}{V_{in}} \right) \\ &= -12.5 \times \left(\frac{50}{50 + 1/g_{m2}} \right) \\ &= -12.5 \times \left(\frac{50}{50 + 40} \right) \\ &= -6.9 \text{ (not too bad)} \end{aligned}$$

Let's drive a CG with a source follower:-



$$\frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_x} \cdot \frac{V_x}{V_{in}}$$

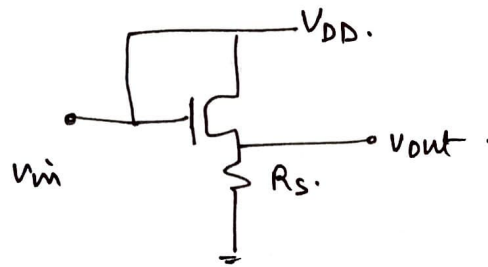
$$\frac{V_x}{V_{in}} = \frac{R_s \parallel 1/g_{m2}}{(R_s \parallel 1/g_{m2}) + 1/g_{m1}}$$

$$\frac{V_{out}}{V_x} = g_{m2} R_D$$

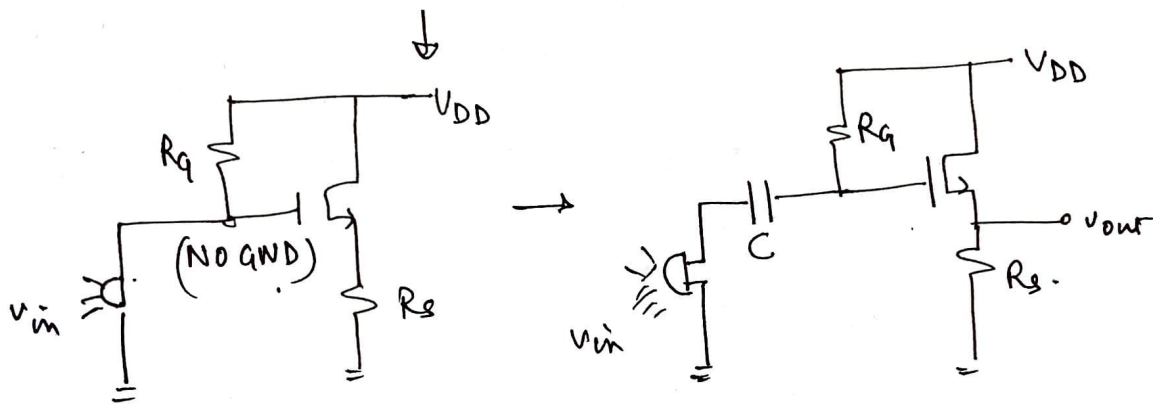
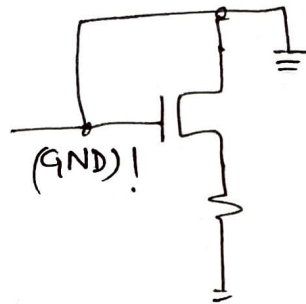
$$\therefore \frac{V_{out}}{V_{in}} = \left(\frac{R_s \parallel 1/g_{m2}}{(R_s \parallel 1/g_{m2}) + 1/g_{m1}} \right) \times g_{m2} R_D$$

Bias design:-

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ac small signal leads to.



Note $\frac{V_{out}}{V_{in}} = \frac{R_E}{r_{gm} + R_E}$ $\therefore$ to increase R_E use a current source.

